23A01604b • OPEN CHANNEL FLOW

UNIT – 1

Hydraulics of Pipelines and Pipe Networks

Complete Study Notes • Concepts · Derivations · Formulae · Exam Questions

Syllabus Topics

- Review of Fluid Mechanics
- Reynolds Transport Theorem and Applications
- Steady Flow Analysis of Pipe Network Systems
- Unsteady Flows – Basic Equations of Water Hammer
- Solution by Method of Characteristics
- Network Analysis

1. REVIEW OF FLUID MECHANICS

1.1 Fluid properties

A **fluid** is a substance that deforms continuously under any applied shear stress. The properties that matter most in hydraulics are given below.

Property	Definition / Symbol	Value for water (20°C)
Mass density	ρ = mass/volume	998 kg/m^3
Specific weight	$\gamma = \rho g$	9.79 kN/m^3
Dynamic viscosity	μ (N·s/m^2), $\tau = \mu\, du/dy$	1.0×10^{-3} Pa·s
Kinematic viscosity	$\nu = \mu/\rho$	1.0×10^{-6} m^2/s
Bulk modulus	$K = -dp/(dV/V)$	2.2 GPa
Surface tension	σ (N/m)	0.073 N/m
Vapour pressure	p_v	2.34 kPa (abs)

Newton's law of viscosity: shear stress is proportional to velocity gradient, **$\tau = \mu\,(du/dy)$**. Fluids obeying this are called **Newtonian**.

1.2 Types of flow

- **Steady / unsteady:** properties at a point do not / do change with time ($\partial/\partial t = 0$ for steady).
- **Uniform / non-uniform:** velocity does not / does change along the flow direction.
- **Laminar / turbulent:** governed by Reynolds number $Re = \rho VD/\mu = VD/\nu$. For pipes, laminar if $Re < 2000$, turbulent if $Re > 4000$. For open channels, laminar if $Re < 500$ (Re based on hydraulic radius R).
- **Compressible / incompressible:** liquids are generally treated as incompressible, except in water-hammer analysis where compressibility is the whole point.
- **Rotational / irrotational:** presence or absence of fluid-element spin.

1.3 Fundamental equations

(a) Continuity (conservation of mass). For steady incompressible flow in a stream-tube:

$$Q = A_1V_1 = A_2V_2 = \text{constant}$$

For unsteady compressible flow the differential form is $\partial\rho/\partial t + \nabla\cdot(\rho\mathbf{V}) = 0$.

(b) Energy equation (Bernoulli). Along a streamline for steady, incompressible, frictionless flow:

$$p_1/\gamma + V_1^2/2g + z_1 = p_2/\gamma + V_2^2/2g + z_2 + h_L$$

Each term has the dimension of length (metres of fluid): **pressure head**, **velocity head** and **elevation (datum) head**. The sum is the **total head H**; the sum $p/\gamma + z$ is the **piezometric head** (hydraulic grade line, HGL) and H is plotted as the **energy grade line (EGL)**. h_L is the head loss between sections 1 and 2.

(c) Momentum equation. The net external force on a control volume equals the rate of change of momentum:

$$\Sigma F = \rho Q(V_2 - V_1) \text{ (steady flow, one inlet and one outlet)}$$

Used for forces on pipe bends, nozzles, hydraulic jump and many open-channel problems.

1.4 Head loss in pipes

Major (friction) loss – Darcy–Weisbach:

$$h_f = f\,L\,V^2 / (2\,g\,D) = 8\,f\,L\,Q^2 / (\pi^2\,g\,D^5)$$

where f is the Darcy friction factor. For laminar flow f = 64/Re. For turbulent flow f depends on Re and relative roughness ε/D and is obtained from the **Colebrook–White equation**:

$$1/\sqrt{f} = -2\log_{10}\left[\,\varepsilon/(3.7D) + 2.51/(Re\,\sqrt{f})\,\right]$$

The **Moody diagram** is the graphical form of this equation. Explicit approximations such as **Swamee–Jain** are used in computer programs:

$$f = 0.25 / \left[\log_{10}\left(\,\varepsilon/(3.7D) + 5.74/Re^{0.9}\,\right)\right]^2$$

Empirical formula – Hazen–Williams (water only, turbulent, SI units):

$$h_f = 10.67\,L\,Q^{1.852} / (C^{1.852}\,D^{4.87})$$

C = Hazen–Williams coefficient (≈ 140 new PVC/smooth, 100–110 old cast iron). Exponent 1.852 is why this formula is popular in network design: it is simple and explicit.

Minor losses:

$$h_m = K\,V^2 / 2g$$

Fitting	Typical K
Sudden contraction	0.2 – 0.5
Sudden expansion	$(1 - A_1/A_2)^2$
Entrance (sharp-edged)	0.5
Exit into reservoir	1.0
90° elbow	0.9
Gate valve (open)	0.2

★ **Key Point:** Minor losses are not "minor" when the pipe is short. A common check is: if L/D > 1000, neglect minor losses.

1.5 Pipes in series, parallel and branching

- **Series:** Q is the same in every pipe; total head loss = Σh_f.
- **Parallel:** head loss is the same in each branch; $Q = \Sigma Q_i$. Equivalent pipe: $Q = \Sigma\,(h_f/r_i)^{1/n}$.
- **Three-reservoir (branching) problem:** continuity at the junction J ($\Sigma Q = 0$) plus energy equation for every pipe. The piezometric head at J is guessed and corrected until the flows balance.

✎ **Exam Question:** A pipe 300 mm diameter, 500 m long, carries water at 0.12 m^3/s. Take f = 0.02. Find the head loss. *Ans: $V = Q/A = 1.70$ m/s; $h_f = 0.02 \times 500 \times 1.70^2/(2 \times 9.81 \times 0.3) \approx 4.9$ m.*

2. REYNOLDS TRANSPORT THEOREM (RTT) AND APPLICATIONS

2.1 System and control volume

A **system** is a fixed collection of fluid particles (Lagrangian view). A **control volume (CV)** is a fixed region in space through which fluid flows (Eulerian view). The conservation laws (mass, momentum, energy) are naturally stated for a system, but we usually measure quantities in a CV. The RTT converts one view into the other.

2.2 Statement of the theorem

Let B be any extensive property (mass, momentum, energy) and b = dB/dm the corresponding intensive property (per unit mass). Then

$$\mathbf{dB_{sys}/dt = \partial/\partial t \int_{CV} \rho\, b\, dV + \oint_{CS} \rho\, b\, (V\cdot n)\, dA}$$

- **Term 1** (left): time rate of change of B for the system.
- **Term 2:** rate of change of B stored inside the control volume (unsteady term).
- **Term 3:** net flux of B through the control surface (convective term). (**V**·**n**) is positive for outflow and negative for inflow.

★ **Key Point:** For steady flow the unsteady term is zero. For one-dimensional flow with uniform properties across inlet and outlet, the surface integral reduces to $(\rho bVA)_{out} - (\rho bVA)_{in}$.

2.3 Derivation (one-dimensional idea)

Consider a system that occupies the CV at time t and has moved slightly at time t + Δt. $B_{sys}(t) = B_{CV}(t)$. At t + Δt the system equals the CV contents, minus the part that has flowed out through the outlet (region II), plus the part that has flowed in (region I) through the inlet. Therefore

$dB_{sys}/dt = \lim [B_{CV}(t+\Delta t) - B_{CV}(t)]/\Delta t + (\text{B-flux out}) - (\text{B-flux in})$,

which on taking the limit Δt → 0 gives the RTT above.

2.4 Applications of RTT

(i) Conservation of mass: put B = m, b = 1.

$$\mathbf{0 = \partial/\partial t \int_{CV} \rho\, dV + \oint_{CS} \rho\, (V\cdot n)\, dA}$$

For steady incompressible flow: $\Sigma Q_{in} = \Sigma Q_{out}$. At a pipe junction this is the node law used in network analysis.

(ii) Linear momentum: put B = m**V**, b = **V**.

$$\mathbf{\Sigma F = \partial/\partial t \int_{CV} \rho V\, dV + \oint_{CS} \rho V\, (V\cdot n)\, dA}$$

Applications: force on a pipe bend, jet on a vane, thrust on a nozzle, and, importantly, the **pressure surge** relation in water hammer (Section 4).

(iii) Energy equation: put B = E, b = e = u + $V^2/2$ + gz.

$$\mathbf{\dot{Q} - \dot{W}_s = \partial/\partial t \int_{CV} \rho e\, dV + \oint_{CS} \rho(e + p/\rho)(V\cdot n)\, dA}$$

Reduces to Bernoulli's equation with head loss for steady flow: $p_1/\gamma + \alpha_1 V_1^2/2g + z_1 + h_{pump} = p_2/\gamma + \alpha_2 V_2^2/2g + z_2 + h_{turbine} + h_L$.

α is the **kinetic energy correction factor** (α = 2 for laminar pipe flow, ≈ 1.05 for turbulent).

(iv) Angular momentum: B = **r** × m**V**; used for turbines, pumps and sprinklers.

2.5 Worked example – force on a pipe bend

Water (Q = 0.2 m^3/s) flows through a horizontal 90° reducing bend from 300 mm to 200 mm diameter. Inlet pressure 200 kPa.

- $V_1 = 0.2/(\pi \cdot 0.15^2) = 2.83$ m/s; $V_2 = 0.2/(\pi \cdot 0.1^2) = 6.37$ m/s.
- Bernoulli (ignoring loss): $p_2 = 200\,000 + 500(2.83^2 - 6.37^2) = 200\,000 - 16\,280 \approx 183.7$ kPa.
- x-direction: $p_1A_1 - F_x = \rho Q(0 - V_1) \rightarrow F_x = 200\,000 \times 0.0707 + 1000 \times 0.2 \times 2.83 \approx 14.7$ kN.
- y-direction: $F_y - p_2A_2 = \rho QV_2 \rightarrow F_y = 183\,700 \times 0.0314 + 1000 \times 0.2 \times 6.37 \approx 7.04$ kN.
- Resultant $F = \sqrt{(14.7^2 + 7.04^2)} \approx 16.3$ kN. The anchor block must resist this force.

✎ **Exam Question:** State the Reynolds Transport Theorem. Deduce the continuity and momentum equations for a control volume. (10 marks)

3. STEADY FLOW ANALYSIS OF PIPE NETWORK SYSTEMS

3.1 What is a pipe network?

A pipe network is a set of pipes connected at **nodes (junctions)** forming one or more **closed loops**. Typical examples: municipal water-distribution systems, irrigation networks, fire-fighting grids. Reasons for loops: reliability (water can reach a node by more than one path) and balancing of pressure.

3.2 Basic laws for network analysis

Law 1 – Continuity at every node:

$$\Sigma Q_{in} - \Sigma Q_{out} = q_{node} \text{ (external demand/supply)}$$

Law 2 – Energy conservation around every closed loop:

$$\Sigma h_f = 0 \text{ (clockwise losses positive, anticlockwise negative)}$$

Law 3 – Head-discharge relation for each pipe:

$$h_f = r\,Q^n$$

where r is the pipe resistance and n is the flow exponent (n = 2 for Darcy–Weisbach with constant f, n = 1.852 for Hazen–Williams).

Formula	Resistance r	Exponent n
Darcy–Weisbach	$8fL/(\pi^2 g D^5)$	2
Hazen–Williams	$10.67\ L/(C^{1.852}\ D^{4.87})$	1.852
Manning	$10.29\ n^2 L/D^{5.33}$	2

3.3 Number of equations

If a network has P pipes, J junctions and L loops, then $P = J + L - 1$. We have $(J - 1)$ independent node equations and L loop equations; with P unknown flows the system is exactly determined. The equations are non-linear (because of Q^n), so they must be solved iteratively.

3.4 Hardy Cross method (loop-balancing method)

Developed by Hardy Cross (1936). It assumes initial flows that satisfy continuity at every node, then corrects each loop by a flow ΔQ so that Σh_f approaches zero.

Let Q_0 be the assumed flow and ΔQ the correction. Then $Q = Q_0 + \Delta Q$ and

$h_f = r(Q_0 + \Delta Q)^n \approx r\,Q_0^{\,n} + n\,r\,Q_0^{\,n-1}\,\Delta Q$ (binomial expansion, higher terms dropped).

Imposing $\Sigma h_f = 0$ for the loop:

$$\Delta Q = -\,\Sigma\, r\,Q_0|Q_0|^{n-1} \,/\, \Sigma\, n\,r\,|Q_0|^{n-1}$$

For n = 2 and clockwise-positive convention: **$\Delta Q = -\,\Sigma\, rQ|Q| \,/\, (2\,\Sigma\, r|Q|)$**.

Step-by-step procedure

- Step 1: Draw the network, number pipes, nodes and loops; mark known inflows/outflows.
- Step 2: Assume a distribution of flows satisfying continuity at each node. Mark the direction (clockwise +ve, anticlockwise −ve).
- Step 3: Compute r for each pipe from D, L and friction formula.
- Step 4: For each loop compute $\Sigma\, rQ|Q|^{n-1}$ (head-loss sum) and $\Sigma\, n\,r|Q|^{n-1}$.

- Step 5: Compute ΔQ and apply it to all pipes in that loop. A pipe common to two loops receives ΔQ from both loops (with opposite sign for the neighbouring loop).
- Step 6: Repeat until |ΔQ| is smaller than the tolerance (e.g. 0.0001 m^3/s) or $\Sigma h_f \approx 0$.

★ **Key Point:** Hardy Cross does NOT require pressure/heads to be assumed – it works with flows only. Its convergence is first-order (linear) and can be slow for large networks, which is why Newton–Raphson methods are used in software.

3.5 Worked example – two-loop network

Consider a single loop ABCD (r in s^2/m^5, n = 2) with assumed flows (clockwise +ve): AB = +40 L/s, BC = +10 L/s, CD = −20 L/s (i.e. anticlockwise flow from D to C), DA = −30 L/s, and r = 400, 800, 300, 500 respectively.

Pipe	Q (m^3/s)	r	h = rQ	Q		2r	Q	
AB	+0.040	400	+0.640	32.0				
BC	+0.010	800	+0.080	16.0				
CD	−0.020	300	−0.120	12.0				
DA	−0.030	500	−0.450	30.0				

Σh = +0.150 m; Σ2r|Q| = 90.0. $\Delta Q = -0.150/90.0 = -0.00167$ m^3/s (anticlockwise correction). New flows: AB = 0.03833, BC = 0.00833, CD = −0.02167, DA = −0.03167. Repeat; after 2–3 iterations Σh → 0.

3.6 Method of nodal heads (H-equations)

Unknowns are the heads H_j at the nodes. For each node the continuity equation is written with $Q_{ij} = ((H_i - H_j)/r_{ij})^{1/n}$. The resulting set of J non-linear equations is solved by **Newton–Raphson**. This is the method used in EPANET and WaterGEMS.

3.7 Method of linear theory (Wood & Charles) and Newton–Raphson for Q-equations

- **Linear theory:** write $h_f = r\,Q_0^{n-1} \cdot Q = r' Q$, i.e. h_f is linear in Q with $r' = r|Q_0|^{n-1}$. Solve the linear system simultaneously, average the new and old flows, and repeat.
- **Simultaneous loop corrections (Newton–Raphson):** solve for all ΔQ_{loop} at once; converges quadratically, but needs matrix inversion.

3.8 Comparison of methods

Method	Unknowns	Convergence	Remark
Hardy Cross (loop)	ΔQ per loop	Linear, slow	Hand calculation; needs initial feasible flows
Newton–Raphson (node heads)	Node heads	Quadratic	Basis of EPANET; robust
Linear theory	Q in all pipes	Fast	Needs no initial feasible flows
Gradient method (Todini)	Q and H together	Quadratic	Handles pumps, valves, PRVs

✎ **Exam Question:** Explain the Hardy Cross method for analysing a water-distribution network and derive the expression for the flow correction ΔQ. (10 marks)

4. UNSTEADY FLOW IN PIPES – WATER HAMMER

4.1 Definition

Water hammer (hydraulic transient, pressure surge) is the series of pressure waves set up in a pressurised pipeline when the flow velocity is changed rapidly, e.g. by sudden valve closure, pump trip or turbine load rejection. Compressibility of water and elasticity of pipe wall, usually neglected in steady flow, become essential.

Causes: valve closure/opening, pump start-up or failure, turbine load rejection, air release, column separation and collapse of voids.

Consequences: pressures of several times static pressure → pipe rupture, joint failure, buckling by sub-atmospheric pressure, vibrations, noise.

4.2 Physical description (valve closed suddenly at downstream end)

Consider a pipeline of length L from a reservoir to a valve, steady velocity V_0. The valve closes instantly at t = 0.

- **0 < t < L/a:** The layer of fluid next to the valve stops. Its kinetic energy converts to elastic strain energy: pressure rises by ΔH and a pressure wave of celerity a travels upstream. Fluid ahead of the wave still has velocity V_0; behind it V = 0.
- **t = L/a:** the wave reaches the reservoir. The whole pipe is at pressure H_0 + ΔH, with V = 0.
- **L/a < t < 2L/a:** The pipe is higher in pressure than reservoir, so fluid flows back toward the reservoir; pressure restores to H_0 as a wave travels downstream with V = $-V_0$.
- **t = 2L/a:** The wave reaches the valve with V = $-V_0$ and p = p_0.
- **2L/a < t < 3L/a:** The fluid at the valve cannot continue to move away; a negative pressure wave (−ΔH) travels upstream, bringing V to 0.
- **3L/a < t < 4L/a:** Pressure returns to normal by the wave moving toward the valve. At t = 4L/a the cycle repeats.

Period of oscillation = 4L/a. In real pipes friction damps the oscillations.

4.3 Joukowsky (Allievi) equation

Apply the RTT momentum equation to a control volume moving with the wave front. Fluid of density ρ approaches the front at relative speed (a + V_0), leaves at (a + V). Neglecting small terms:

$$\Delta H = -(a/g)\,\Delta V \quad \text{or} \quad \Delta p = -\rho\, a\, \Delta V$$

For complete stoppage ΔV = $-V_0$, so $\Delta H = a\,V_0/g$.

Example: a = 1000 m/s, V_0 = 2 m/s → ΔH = 1000 × 2/9.81 ≈ 204 m (≈ 2.0 MPa). This is why rapid valve closure is dangerous.

4.4 Wave speed (celerity)

$$a = \sqrt{(K/\rho)} \,/\, \sqrt{(1 + (K\,D / E\,e)\,c_1)}$$

- K = bulk modulus of the liquid; ρ = density; D = pipe diameter; e = wall thickness; E = Young's modulus of pipe material.
- c_1 is the pipe-support factor: for thin-walled pipe anchored at upstream end $c_1 = 1 - \mu_p/2$; anchored throughout against longitudinal movement $c_1 = 1 - \mu_p^2$ (μ_p = Poisson's ratio).
- For a rigid pipe (E → ∞): a = √(K/ρ) ≈ 1450 m/s for water.

Pipe material	Typical wave speed a (m/s)
Rigid (theoretical)	1450
Steel	1000 – 1250
Concrete / cast iron	1000 – 1200
Asbestos cement	900 – 1100
PVC / PE	200 – 500

4.5 Classification of valve closure

- **Critical period** T_c = 2L/a, the time taken for the pressure wave to travel to the reservoir and return.
- **Rapid (sudden) closure:** $T_{closure} \leq 2L/a$. The full Joukowsky rise occurs: $\Delta H = aV_0/g$.
- **Slow closure:** $T_{closure} > 2L/a$. Reflected relief wave arrives before closure is finished, so the maximum rise is smaller. Approximate (Michaud): $\Delta H \approx 2\ L\ V_0\ /(g\ T_{closure})$.

★ **Key Point:** Closing the valve in a time > 2L/a does not eliminate transients, but reduces them significantly. Surge tanks, air vessels, pressure-relief valves and flywheels are used to control them.

✎ **Exam Question:** A steel pipe, 1.2 m diameter, 8 mm thick, 1500 m long carries water at 2.5 m/s. Compute the pressure rise for a valve closure in 1.0 s. (Use a = 1000 m/s.) *Ans: 2L/a = 3 s; closure 1 s < 3 s → rapid. ΔH = 1000 × 2.5/9.81 ≈ 255 m.*

5. BASIC EQUATIONS OF WATER HAMMER

5.1 Assumptions

- Flow is one-dimensional; velocity V(x,t) and piezometric head H(x,t) are cross-section averages.
- Pipe is full, linear-elastic and prismatic; wave speed a is constant.
- Liquid is slightly compressible; density variation is small except where multiplied by a.
- Unsteady friction is represented by the steady-state Darcy–Weisbach term.
- Convective terms V∂V/∂x and V∂H/∂x are small compared with other terms because V ≪ a.

5.2 Continuity equation

Apply the mass conservation to an elementary pipe length δx. The mass accumulates due to (i) liquid compression and (ii) pipe-wall expansion, both functions of pressure change. This gives

$$\partial H/\partial t + (a^2/g)\,\partial V/\partial x + V\,\partial H/\partial x + V \sin\theta = 0 \rightarrow \partial H/\partial t + (a^2/g)\,\partial V/\partial x = 0 \text{ (simplified)}$$

Here θ is the pipe inclination.

5.3 Momentum (dynamic) equation

Apply Newton's second law to the fluid element of length δx: pressure gradient + gravity + wall shear = mass × acceleration.

$$\partial V/\partial t + g\,\partial H/\partial x + (f\,V|V|)/(2D) = 0$$

Here the last term is the wall friction (Darcy–Weisbach form, with |V| to retain direction).

5.4 Summary – the two governing PDEs

	Form		
	$\partial H/\partial t + (a^2/g)\,\partial V/\partial x = 0$		
	$\partial V/\partial t + g\,\partial H/\partial x + f\,V$	V	/(2D) = 0

They are a pair of coupled, quasi-linear, hyperbolic partial differential equations with two unknowns H(x,t) and V(x,t). Analytical solutions exist only for very idealised cases (frictionless, linear), so numerical solution – the **method of characteristics (MOC)** – is used.

5.5 Rigid-column theory vs elastic theory

- **Rigid-column theory:** neglects compressibility; the entire liquid column accelerates as one body. Governed by L/g dV/dt = ΔH. Valid for slow transients (gradual valve operation, surge tanks, mass oscillation).
- **Elastic (water hammer) theory:** includes compressibility and wall elasticity. Valid for fast transients where pressure waves travel ($T_{closure} \lesssim 10$ L/a).

6. SOLUTION BY THE METHOD OF CHARACTERISTICS (MOC)

6.1 Idea

MOC converts the two PDEs into four ordinary differential equations valid along special lines (characteristic lines) in the x–t plane. These ODEs are integrated by finite differences along the characteristics. It handles non-linear friction, complex boundary conditions (valves, pumps, reservoirs, surge tanks) and is the industry standard (e.g. Wylie & Streeter).

6.2 Derivation

Let $L_1 = \partial V/\partial t + g\,\partial H/\partial x + (f\,V|V|)/(2D) = 0$ (momentum)

Let $L_2 = (g/a^2)\,\partial H/\partial t + \partial V/\partial x = 0$ (continuity)

Form the linear combination $L = L_1 + \lambda L_2$:

$(\partial V/\partial t + \lambda\,\partial V/\partial x) + \lambda\,(g/a^2)\,(\partial H/\partial t + (a^2/\lambda)\,\partial H/\partial x) + f\,V|V|/(2D) = 0.$

Choose λ so that dx/dt is the same for V and H: $dx/dt = \lambda$ and $dx/dt = a^2/\lambda$, hence $\lambda = \pm a$. Then the bracketed terms become total derivatives dV/dt and dH/dt:

$$\mathbf{C^+: (g/a)\,dH/dt + dV/dt + f\,V|V|/(2D) = 0 \text{ valid along } dx/dt = +a}$$

$$\mathbf{C^-: -(g/a)\,dH/dt + dV/dt + f\,V|V|/(2D) = 0 \text{ valid along } dx/dt = -a}$$

6.3 Finite-difference form

Divide the pipe into N equal reaches $\Delta x = L/N$ and choose $\Delta t = \Delta x/a$ (Courant condition, so characteristics pass exactly through grid points). To find conditions at point P (position i, time t + Δt) use points A (i−1, t) and B (i+1, t):

$$\mathbf{C^+: H_P = C_P - B_P\,Q_P \text{ where } C_P = H_A + B\,Q_A - R\,Q_A|Q_A|}$$

$$\mathbf{C^-: H_P = C_M + B_M\,Q_P \text{ where } C_M = H_B - B\,Q_B + R\,Q_B|Q_B|}$$

Here $B = a/(gA)$ is the **pipe characteristic impedance**, $R = f\,\Delta x/(2\,g\,D\,A^2)$ is the **resistance coefficient**, and Q is discharge. Solve the two equations for the two unknowns (Q_P, H_P) at interior nodes:

$$\mathbf{Q_P = (C_P - C_M)/(B_P + B_M),\ H_P = (C_M\,B_P + C_P\,B_M)/(B_P + B_M)}$$

6.4 Boundary conditions

- **Upstream reservoir (constant head H_R):** $H_P = H_R$; only C^- is available → $Q_P = (H_R - C_M)/B_M$.
- **Downstream dead end / closed valve:** $Q_P = 0$; only C^+ → $H_P = C_P$.
- **Downstream valve (orifice):** $Q_P = (C_d A_G)_P\,\sqrt{(2g\,H_P)}$; combine with C^+ to get a quadratic in Q_P.
- **Series junction (change of diameter):** continuity $Q_{P1} = Q_{P2}$ and equal head (neglecting losses); two C^+, C^- equations.
- **Branching junction:** $\Sigma Q = 0$, equal head H at the node; N + 1 unknowns with N characteristics.
- **Pump:** head–discharge curve $H_{pump} = a_0 + a_1 Q + a_2 Q^2$ combined with the characteristic.
- **Surge tank:** continuity of tank level z_s + equal heads.

6.5 Computational steps

- 1. Compute steady-state initial values of H and Q at all nodes (t = 0).

- 2. Select N, $\Delta x = L/N$, $\Delta t = \Delta x/a$.
- 3. For each time step, compute interior nodes using C^+ and C^-.
- 4. Apply the boundary conditions at the two ends.
- 5. Store H and Q; advance $t = t + \Delta t$ until the required duration is reached.
- 6. Plot H versus t at critical points (valve, mid-point) and envelope of maximum/minimum head along the pipe.

★ **Key Point:** If pipes have different wave speeds, the grid spacing is adjusted so that Δt is common to all pipes ($\Delta x_i = a_i \Delta t$), or $\Delta t = \Delta x/a$ is slightly altered ("wave-speed adjustment" interpolation).

6.6 Worked example – frictionless line

Reservoir head 100 m, L = 600 m, a = 1200 m/s, V_0 = 1.5 m/s, valve closed instantly. N = 2, Δx = 300 m, Δt = 0.25 s. At t = 0.25 s: valve node $H = 100 + a V_0/g = 100 + 183.5 = 283.5$ m; at middle node H = 100 m; V = 0 at valve. At 0.5 s: wave has reached reservoir; the middle node reaches 283.5 m. At 1.0 s (= 2L/a) the pipe is at rest, high pressure. At t = 1.0–2.0 s the wave reverses. A plot of the valve head vs. time is a **square wave** between 283.5 m and −83.5 m (limited to vapour pressure, causing column separation) of period 4L/a = 2.0 s.

✎ **Exam Question:** Derive the characteristic equations of water hammer and explain the finite-difference solution procedure. (15 marks)

7. WATER HAMMER CONTROL AND PROTECTION

7.1 Methods of controlling surges

- **Slow valve operation:** keep closure time T_c > 2L/a, preferably > 5 L/a.
- **Surge tank (surge shaft):** an open standpipe connected near the turbine or valve. It reflects the pressure wave and provides/absorbs water. Types: simple, restricted orifice, differential (Johnson), closed (air) surge tank.
- **Air vessel (hydropneumatic accumulator):** a closed vessel with compressed air cushion; effective at pumping stations.
- **One-way surge tank / feed tank:** supplies water only when the pipe pressure drops below tank level to prevent column separation.
- **Pressure-relief / surge-anticipation valves:** open at preset pressure and release flow.
- **Air-inlet and vacuum-breaker valves** at high points to avoid sub-atmospheric pressure and pipe collapse.
- **Flywheel on pump:** increases inertia so pump coasts down slowly after power failure.
- **Bypass line with non-return valve** around the pump.

7.2 Column separation

When pressure falls to the vapour pressure of water, vapour cavities form and the liquid column separates. Subsequent collapse of the cavity causes a sharp pressure spike that can exceed the Joukowsky value. It is modelled by the **discrete vapour cavity model (DVCM)** within MOC.

7.3 Pipe design for surge

Pipe pressure class must exceed (static head + surge head). Required wall thickness from the hoop-stress equation: $e = p\,D / (2\,\sigma_{allow})$. The wave speed a decreases with decreasing wall thickness, so thin pipes have smaller surge pressures but must still resist the total pressure.

8. NETWORK ANALYSIS – MODERN COMPUTER APPROACH

8.1 Steady-state analysis in software (EPANET)

Computer programs treat the network as a graph (links = pipes, pumps, valves; nodes = junctions, reservoirs, tanks). Input: pipe lengths, diameters, roughness, node demands, elevations, pump curves. Output: flows, velocities, pressures at nodes.

Gradient algorithm: solves the energy equation for all links and the continuity equation for all nodes simultaneously through Newton–Raphson iterations, linearised each time:

$$\mathbf{A_{11}\,Q + A_{12}\,H = -A_{10}\,H_0,\ A_{21}\,Q = q}$$

where A_{12} is the connectivity matrix (incidence), A_{11} is a diagonal matrix of head-loss gradients $n\,r|Q|^{n-1}$, and q is the demand vector.

8.2 Extended-period simulation (EPS)

Demand patterns, tank filling/draining, pump on–off controls simulated through time using successive steady-state analyses. Useful for planning storage and pumping schedules.

8.3 Design criteria for distribution networks

- Minimum residual pressure at nodes: 7–15 m for ordinary domestic supply (≥ 17 m for 3-storey building).
- Maximum velocity ≈ 1.5–2.0 m/s; minimum ≈ 0.3 m/s to avoid sedimentation.
- Loops of equal head-loss; no dead-ends where possible.
- Fire flow demand (e.g. 100 L/s for 4 hr) to be checked.

8.4 Transient network analysis

Surge analysis of branched/looped networks uses the MOC at every pipe, with junction compatibility equations at nodes. Software: HAMMER, SURGE, ALLIEVI, TSNet. Steps: (a) steady state initialisation, (b) event definition (pump trip, valve closure), (c) time-marching MOC, (d) envelope of max/min heads, (e) design of protection devices, (f) re-simulation.

8.5 Equivalent pipe concept and Dupuit equation

Two pipes are equivalent if the same discharge gives the same head loss. For a series system: $L_e/D_e^{4.87} = \Sigma\, L_i/D_i^{4.87}$ (H–W, with same C). For Dupuit equation in the parallel case: $Q = \Sigma\, Q_i$, and with $D_e^{2.63}/\sqrt{L_e^{0.54}} = \Sigma\, D_i^{2.63}/L_i^{0.54}$.

8.6 Quick revision – formulae sheet

	Q	$^{n-1}$ / Σ n r	Q

8.7 Important short questions

- Define water hammer. What are its causes?

- State the Joukowsky equation and give its limitation.
- Why is the Courant condition $\Delta t = \Delta x/a$ important in MOC?
- Distinguish between rigid column theory and elastic theory.
- List four methods of controlling surge pressure.
- Why is Hardy Cross method called a loop method? Write the condition at each node and loop.

23A01604b • OPEN CHANNEL FLOW

UNIT – 2

Steady Varied Flows in Open Channels

Complete Study Notes • Concepts · Derivations · Formulae · Exam Questions

Syllabus Topics

- Basic Concepts of Uniform Flow
- Specific Energy and Specific Force Concepts
- Dynamic Equation for Spatially Varied Flows
- Flow Profile Computations
- Introduction to HEC-RAS
- Spatially Varied Flows and Rapidly Varied Flows – Applications

1. BASIC CONCEPTS OF OPEN CHANNEL FLOW

1.1 Open channel vs pipe flow

An **open channel** is a conduit in which a liquid flows with a **free surface** exposed to atmospheric pressure. The driving force is gravity (component of weight along the bed slope), whereas in pipes it is the pressure gradient.

Feature	Pipe flow	Open-channel flow
Driving force	Pressure gradient	Gravity (bed slope)
Free surface	Absent (flows full)	Present; p = 0 gauge
Flow section	Fixed geometry	Depth varies with Q, so section varies
Roughness	Fairly constant	Varies with depth, bed forms, vegetation
Analysis	Easier	Harder: unknown depth is part of the solution

1.2 Channel geometry

- **Prismatic channel:** constant cross-section and bed slope. **Non-prismatic:** natural rivers.
- **Depth y:** vertical distance from the lowest point of the section to the free surface. **Flow depth normal to bed** $d = y \cos\theta$.
- **Top width T:** width at free surface. **Area A. Wetted perimeter P. Hydraulic radius R = A/P. Hydraulic depth D = A/T.**
- **Section factor** for critical flow $Z = A\sqrt{D} = A\sqrt{(A/T)}$; for uniform flow $AR^{2/3}$.

Section	Area A	Wetted perimeter P	Top width T
Rectangle (b, y)	by	b + 2y	b
Trapezoid (b, y, side slope m:1)	(b + my)y	$b + 2y\sqrt{(1 + m^2)}$	b + 2my
Triangle (m, y)	my^2	$2y\sqrt{(1 + m^2)}$	2my
Circle (D, central angle θ)	$(D^2/8)(\theta - \sin\theta)$	Dθ/2	$D \sin(\theta/2)$
Parabola	(2/3)Ty	$T + 8y^2/3T$	—

1.3 Classification of open-channel flow

- **Steady vs unsteady** (time).
- **Uniform vs varied** (space). Varied flow is subdivided into **gradually varied flow (GVF)** where depth changes slowly over long distances (e.g. backwater curves), and **rapidly varied flow (RVF)** where depth changes abruptly over short distance (hydraulic jump, flow over a weir, sluice-gate flow).
- **Spatially varied flow (SVF):** discharge changes along the channel because of lateral inflow or outflow (side-weir, gutter, wash-water trough).
- **Subcritical, critical and supercritical:** depends on Froude number

$$\mathbf{Fr = V / \sqrt{(g\,D)} = V / \sqrt{(g\,A/T)}}$$

Fr < 1 subcritical (tranquil), Fr = 1 critical, Fr > 1 supercritical (rapid). Information (small surface waves of speed $c = \sqrt{(gD)}$) can travel upstream in subcritical flow only; hence **subcritical flow is controlled from downstream and supercritical flow from upstream**.

- **Laminar vs turbulent:** Re = VR/ν; laminar < 500, turbulent > 2000 (practically always turbulent).

1.4 Pressure distribution and velocity distribution

Pressure is **hydrostatic** for channels of small slope and parallel streamlines: $p = \gamma(y - z)$. The velocity varies across the section; maximum velocity occurs slightly below the free surface (about 0.05–0.25 y). The mean velocity is approximately the velocity at 0.6 y from the surface, or the average of velocities at 0.2 y and 0.8 y. **Energy correction factor α** = $(1/AV^3)\int v^3 dA \approx 1.03–1.36$; **momentum correction β** ≈ 1.01–1.12. Usually taken as 1.

2. UNIFORM FLOW

2.1 Definition and conditions

Uniform flow occurs when depth, area, velocity and discharge remain constant along the channel. Then the bed slope S_0, water surface slope S_w and energy-line slope S_f are all equal:

$$\mathbf{S_0 = S_w = S_f}$$

The flow depth is called **normal depth y_n**. It is reached only in long, prismatic, straight channels of constant roughness when gravity force equals boundary shear force.

Force balance on a control volume: $\gamma A L \sin\theta = \tau_0 P L \rightarrow \tau_0 = \gamma R S_0$.

2.2 Chezy equation

Equating shear stress $\tau_0 = k \rho V^2$ to $\gamma R S_0$ gives

$$\mathbf{V = C\sqrt{(R S_0)}}$$

C = Chezy's coefficient ($m^{1/2}/s$). Empirical: **Kutter** and **Bazin** formulae; $C = (1/n)R^{1/6}$ (Manning); $C = \sqrt{(8g/f)}$ (Darcy–Weisbach).

2.3 Manning's equation

The most widely used formula in SI units:

$$\mathbf{V = (1/n)\, R^{2/3}\, S_0^{1/2} \text{ and } Q = (1/n)\, A\, R^{2/3}\, S_0^{1/2}}$$

Surface	Manning n
Smooth concrete	0.012 – 0.014
Brick / rough concrete	0.015 – 0.017
Earth channel, clean	0.018 – 0.025
Earth with weeds	0.030 – 0.040
Natural stream, clean, straight	0.025 – 0.033
Mountain stream with boulders	0.040 – 0.070
Flood plain with trees	0.075 – 0.150

Conveyance $K = (1/n) A R^{2/3}$ so that $Q = K\sqrt{S_0}$.

2.3.1 Computation of normal depth

Given Q, n, S_0 and geometry, solve $A R^{2/3} = Q n/\sqrt{S_0}$ for y_n. This is non-linear, so use trial-and-error, charts, Newton–Raphson or software. For a **wide rectangular channel** $R \approx y$: $q = (1/n) y^{5/3} S_0^{1/2} \rightarrow y_n = (qn/\sqrt{S_0})^{3/5}$.

✎ **Exam Question:** A trapezoidal channel with b = 5 m, m = 1.5, n = 0.025, S_0 = 0.0004 carries 12 m^3/s. Find y_n. *Method: required $AR^{2/3} = 12 \times 0.025/0.02 = 15.0$; trial y = 1.5 m → A = 10.875, P = 10.41, R = 1.045, $AR^{2/3}$ = 11.2; y = 1.8 m → A = 13.86, P = 11.49, R = 1.206, $AR^{2/3}$ = 15.8; y ≈ 1.75 m.*

2.4 Most efficient (best hydraulic) section

For a given area A and slope, discharge is maximum when wetted perimeter P is minimum.

- **Rectangle:** b = 2y, R = y/2.

- **Trapezoid:** half of a regular hexagon, side slope 60° (m = 1/√3 = 0.577); R = y/2 for every efficient section.
- **Triangle:** side slope 45° (m = 1), R = y/(2√2).
- **Circle:** maximum velocity at y = 0.81 D; maximum discharge at y = 0.95 D.

★ **Key Point:** Best hydraulic section is not always the most economical: excavation, lining and construction practice must also be considered.

2.5 Compound channels

Divide the section into sub-areas (main channel + flood plains) by vertical lines (not counted in wetted perimeter), compute Q for each with its own n, and sum: $Q = \Sigma K_i \sqrt{S_0}$. Equivalent roughness (Horton/Einstein): $n_e = [\Sigma P_i n_i^{1.5}/P]^{2/3}$.

2.6 Design of non-erodible channels

Steps: (1) choose n for lining, (2) choose side slope m, (3) check maximum permissible velocity, (4) assume b/y ratio, (5) compute A from Q/V, (6) check Froude number, (7) add **freeboard** (typically 0.3–0.9 m, or 5–30% of depth).

3. SPECIFIC ENERGY

3.1 Definition

Specific energy E is the energy per unit weight of water measured with respect to the channel bed as datum:

$$E = y + \alpha V^2/2g = y + Q^2/(2\,g\,A^2)$$

For a rectangular channel with discharge per unit width q = Q/b: $\mathbf{E = y + q^2/(2gy^2)}$.

3.2 Specific energy curve (for constant q)

E is a cubic in y: $y^3 - E\,y^2 + q^2/2g = 0$. For a given E there are generally three roots; one negative (meaningless) and two positive roots called **alternate depths** y_1 (supercritical, small) and y_2 (subcritical, large). At the minimum energy E_{min}, the two depths coincide at **critical depth y_c**.

- Upper limb of the curve: subcritical ($y > y_c$); lower limb: supercritical ($y < y_c$).
- Asymptotes: E = y (45° line) and y = 0 (the E axis).

3.3 Critical flow conditions

Set $dE/dy = 0$: $1 - Q^2T/(gA^3) = 0$, hence

$$Q^2/g = A^3/T \Leftrightarrow Fr = 1$$

- **Rectangular:** $y_c = (q^2/g)^{1/3}$; $V_c = \sqrt{(g\,y_c)}$; $\mathbf{E_{min} = (3/2)\,y_c}$; $V_c^2/2g = y_c/2$.
- **Triangular:** $y_c = (2Q^2/(g\,m^2))^{1/5}$; $E_{min} = (5/4)y_c$.
- **Trapezoidal:** solve by trial: $Q^2/g = [(b + my_c)y_c]^3/(b + 2my_c)$.
- **Circular:** use charts or trial.

★ **Key Point:** Critical depth depends ONLY on discharge and channel geometry (not on slope or roughness). Normal depth depends on both. When $y_n > y_c$ the slope is **mild**; $y_n < y_c$: **steep**; $y_n = y_c$: **critical** slope S_c; $y_n \to \infty$: **adverse**; $S_0 = 0$: **horizontal**.

3.4 Alternate depths and transitions

Hump (raised bed Δz): $E_2 = E_1 - \Delta z$. For subcritical approach, y drops over the hump; for supercritical approach, y rises. If $\Delta z > E_1 - E_{min}$, **choking** occurs: upstream depth rises (subcritical) to supply the required energy and critical flow occurs on the hump crest.

Width contraction (rectangular): q increases from q_1 to $q_2 = Q/b_2$. Maximum contraction for no change upstream: $b_{2,min} = Q/[\sqrt{g}\,(2E_1/3)^{3/2}]$ (i.e. condition $E_1 = E_{min}$ at the throat).

3.5 Worked example

q = 3.0 m^2/s, y_1 = 2.0 m. Find alternate depth. V_1 = 1.5 m/s, $E_1 = 2.0 + 1.5^2/19.62 = 2.115$ m. $y_c = (9/9.81)^{1/3} = 0.972$ m. Solve $y + 0.4587/y^2 = 2.115$ for the small root by trial: y = 0.55 gives 2.066; y = 0.54 gives 2.113. So the alternate (supercritical) depth is $\mathbf{y_2 \approx 0.54\ m}$ (V = 5.56 m/s, Fr = 2.4).

4. SPECIFIC FORCE (MOMENTUM FUNCTION)

4.1 Concept

When energy losses are unknown (e.g. hydraulic jump) the **momentum equation** is used. The **specific force** (force per unit weight of water) at a section is

$$\mathbf{M = Q^2/(gA) + A\,\bar{y}}$$

where $A\,\bar{y}$ is the first moment of the area about the free surface ($\bar{y}$ = depth of centroid below the water surface) and Q^2/gA is the momentum flux. For a rectangular channel: $\mathbf{M = q^2/(g\,y) + y^2/2}$ per unit width.

4.2 Specific force curve

Plotting y against M for a constant Q gives a curve with a minimum M at **critical depth** ($dM/dy = 0$ gives the same condition $Q^2/g = A^3/T$). For a given M two depths exist called **conjugate (sequent) depths** y_1 and y_2.

Concept	Energy (specific energy)	Momentum (specific force)
Quantity conserved	Energy (frictionless)	Force/momentum
Depth pair	Alternate depths (E same)	Conjugate depths (M same)
Use	Smooth transitions, weirs	Hydraulic jump, flow through gates, forces on structures
Minimum at	Critical depth	Critical depth

4.3 Hydraulic jump (introductory)

A **hydraulic jump** is the abrupt transition from supercritical to subcritical flow with large energy dissipation; the momentum equation gives ($M_1 = M_2$):

$$\mathbf{y_2/y_1 = \tfrac{1}{2}\,[\,\sqrt{(1 + 8\,Fr_1^{\,2})} - 1\,]\ (rectangular\ channel)}$$

Energy loss: $\mathbf{h_L = (y_2 - y_1)^3/(4\,y_1\,y_2)}$. Length of jump $L_j \approx 6\,y_2$ (5–7). Efficiency = E_2/E_1.

Fr_1	Type of jump	Energy dissipation
1.0 – 1.7	Undular	< 5%
1.7 – 2.5	Weak	5 – 15%
2.5 – 4.5	Oscillating	15 – 45%
4.5 – 9.0	Steady (best)	45 – 70%
> 9.0	Strong (choppy)	70 – 85%

4.4 Applications of specific force

- Stilling-basin design for dams and spillways.
- Jump below sluice gates; forces on gate: $F = \gamma(M_1 - M_2)$ shifted.
- Location of jump on sloping apron – use y_2 from jump equation vs tailwater depth: tailwater > y_2 drowns the jump; < y_2 sweeps it downstream.

✎ **Exam Question:** A rectangular channel carries 5 $m^3/s/m$ at $y_1 = 0.5$ m. Determine y_2, loss of head and power dissipated. *Ans: $V_1 = 10$ m/s, $Fr_1 = 4.52$, $y_2 = (0.5/2) \times (\sqrt{(1 + 8\times20.4)} - 1) = 2.95$ m, $h_L = (2.45)^3/(4\times0.5\times2.95) = 2.49$ m, $P = \gamma\,q\,h_L = 9.81 \times 5 \times 2.49 \approx 122$ kW per metre width.*

5. GRADUALLY VARIED FLOW (GVF) – THE DYNAMIC EQUATION

5.1 Assumptions

- Flow is steady; streamlines are nearly parallel (hydrostatic pressure).
- Bed slope is small ($\cos\theta \approx 1$, $\sin\theta \approx \tan\theta = S_0$).
- Channel is prismatic with constant α and roughness; head loss computed from uniform-flow formula using local V and R: $S_f = n^2V^2/R^{4/3}$.
- Discharge is constant along the reach (for SVF see Section 8).

5.2 Derivation

Total head $H = z + y + \alpha V^2/2g$. Differentiate with respect to x (distance along bed): $dH/dx = -S_f$ (energy slope is the negative of H gradient). Also $dz/dx = -S_0$. So

$-S_f = -S_0 + dy/dx + d/dx\,(\alpha Q^2/2gA^2)$.

Since $dA = T\,dy$, $d/dx(Q^2/2gA^2) = -(Q^2T/gA^3)\,dy/dx = -Fr^2\,dy/dx$. Therefore:

$$\mathbf{dy/dx = (S_0 - S_f) / (1 - Fr^2)}$$

This is the **dynamic equation of GVF**. Alternative forms with Manning's formula, using K = conveyance and K_n = normal conveyance, $Z = A\sqrt{D}$:

$$\mathbf{dy/dx = S_0\,[1 - (K_n/K)^2] / [1 - (Z_c/Z)^2]}$$

For a wide rectangular channel ($R \approx y$): $\mathbf{dy/dx = S_0\,[1 - (y_n/y)^{10/3}] / [1 - (y_c/y)^3]}$.

5.3 Interpretation of the equation

- Numerator > 0 when $y > y_n$ ($S_f < S_0$); < 0 when $y < y_n$.
- Denominator > 0 when $y > y_c$ (Fr < 1); < 0 when $y < y_c$.
- $dy/dx > 0$ means depth increases downstream. As $y \to y_n$, $dy/dx \to 0$ (flow becomes uniform – the profile is asymptotic to the normal depth line). As $y \to y_c$, $dy/dx \to \infty$ (the profile meets the critical depth line at a steep angle, and the GVF assumption breaks down).

6. CLASSIFICATION OF WATER-SURFACE PROFILES

6.1 Zones and slope types

Draw the normal depth line (NDL) and critical depth line (CDL). They divide the space above the bed into three zones:

- **Zone 1:** y above both lines.
- **Zone 2:** y between the lines.
- **Zone 3:** y below both lines.

Slopes: **M** (mild, $y_n > y_c$), **S** (steep, $y_n < y_c$), **C** (critical, $y_n = y_c$), **H** (horizontal, $S_0 = 0$), **A** (adverse, $S_0 < 0$). Profile name = slope letter + zone number (M1, M2, M3, S1, S2, S3, C1, C3, H2, H3, A2, A3).

6.2 Summary of the 12 profiles

Profile	Condition	dy/dx	Shape	Example
M1	$y > y_n > y_c$	+	Rising, asymptotic to horizontal at downstream end	Backwater upstream of dam
M2	$y_n > y > y_c$	−	Drawdown	Approach to a free overfall / change to steeper slope
M3	$y_n > y_c > y$	+	Rising, supercritical	Downstream of sluice gate on mild slope, ends in jump
S1	$y > y_c > y_n$	+	Rising to horizontal	Upstream of a dam/weir on steep slope; begins with jump
S2	$y_c > y > y_n$	−	Falling to y_n	Mild-to-steep slope break (downstream)
S3	$y_c > y_n > y$	+	Rising to y_n	Below a gate on steep slope
C1 / C3	$y > y_c = y_n$; $y < y_c$	+	Horizontal at y_c	Transition near critical slope
H2 / H3	$S_0 = 0$	− / +	Drawdown / rising	Flat apron / channel end
A2 / A3	$S_0 < 0$	− / +	Drawdown / rising	Adverse slope

★ **Key Point:** Zone 2 profiles are always drawdown curves (depth falls) and zone 1 and 3 are always rising (backwater) curves. Subcritical flow profiles (M1, M2, S1, H2, A2) are computed from the downstream control going upstream; supercritical flow profiles (M3, S2, S3, H3, A3) are computed from the upstream control going downstream.

6.3 Control sections

A **control** is a section where a known depth–discharge relation exists: critical depth at a free overfall or crest of a weir; sluice gate (vena contracta depth); reservoir (known level); normal depth far upstream/downstream.

✎ **Exam Question:** Sketch and explain the possible GVF profiles when a mild slope is followed by a steep slope; and when a steep slope is followed by a mild slope (jump). (8 marks)

7. FLOW PROFILE COMPUTATIONS

7.1 Direct step method (distance from depth)

For prismatic channels. Choose consecutive depths y_1, y_2 and compute the distance Δx. From energy balance between sections:

$$\Delta x = (E_2 - E_1) / (S_0 - S_f)$$

where $E = y + \alpha V^2/2g$ and $S_f = ½(S_{f1} + S_{f2})$ (or the Manning S_f at mean R and V). Steps: choose y_1 at the control; choose Δy; compute E, S_f at both sections; calculate Δx; Σ Δx gives position. Positive Δx in the downstream direction for supercritical (go downstream); negative for subcritical (go upstream). Advantage: no iterations. Limitation: only for prismatic channels; the distances are results, not inputs.

7.2 Standard step method (depth from distance)

Used for natural channels with irregular geometry. The stations x_i are fixed (cross-sections surveyed); depth/water level at the next section is found by trial:

$$z_1 + y_1 + \alpha_1 V_1^2/2g = z_2 + y_2 + \alpha_2 V_2^2/2g + h_f + h_e$$

with $h_f = S_f \Delta x$ and eddy/contraction loss $h_e = C_e|\alpha_2 V_2^2/2g - \alpha_1 V_1^2/2g|$ (C_e = 0.1 contraction, 0.3 expansion). Assume y_2, compute RHS, compare with LHS, correct and iterate until the balance error is < 0.003 m. This is exactly the algorithm in HEC-RAS.

7.3 Numerical integration of the dynamic equation

- **Euler / Predictor-corrector:** $y_{i+1} = y_i + \Delta x\, f(x_i, y_i)$.
- **Runge–Kutta 4th order:** more accurate, larger steps.
- **Bresse's method:** analytical solution for wide rectangular channels using the varied-flow function F(u, N) tables, with $u = y/y_n$:

$$x = (y_n/S_0)\,[\,u - F(u, N) + (y_c/y_n)^3\, F(u, 3)\,] + C$$

- **Chow's method:** uses varied-flow functions with hydraulic exponents N and M for general channels.

7.4 Worked example (direct step)

Rectangular channel b = 10 m, n = 0.015, S_0 = 0.0004, Q = 30 m^3/s. Trial gives $y_n \approx 1.85$ m; $y_c = (3^2/9.81)^{1/3} = 0.97$ m, so the slope is mild. A dam raises depth to 3.0 m (M1 profile). Find the distance to y = 2.5 m.

- At y = 3.0: A = 30, V = 1.00 m/s, R = 1.875, E = 3.051 m, $S_f = (0.015\times1.0)^2/1.875^{4/3} = 9.7\times10^{-5}$.
- At y = 2.5: A = 25, V = 1.20 m/s, R = 1.667, E = 2.573 m, $S_f = 1.64\times10^{-4}$.
- $S_f = 1.31\times10^{-4}$; Δx = (2.573 − 3.051)/(0.0004 − 0.000131) = −1774 m, i.e. the depth is 2.5 m about 1770 m **upstream** of the dam. Repeat for smaller steps; the profile approaches y_n asymptotically.

8. SPATIALLY VARIED FLOW (SVF)

8.1 Definition and types

SVF has non-uniform discharge along the channel, due to addition or withdrawal of water along its length.

- **Increasing discharge (lateral inflow):** roadside gutters, spillway side-channel, wash-water troughs in filters, drainage.
- **Decreasing discharge (lateral outflow):** side weirs, bottom racks (screens), irrigation canal with turnouts, perforated pipe.

8.2 Dynamic equation – increasing discharge

Apply the **momentum equation** (not energy, because mixing of the inflow causes unknown energy loss) to a CV of length dx:

$$\mathbf{dy/dx = [S_0 - S_f - (2Qq^*)/(gA^2)] / (1 - Q^2T/(gA^3)) \text{ where } q^* = dQ/dx}$$

(Assumes the lateral inflow enters at right-angle with no x-momentum.) Used for the side-channel spillway: depth increases downstream in the channel, critical section occurs at the end (control section, depends on friction).

8.3 Dynamic equation – decreasing discharge

Here the diverted water has the same specific energy as the main flow, so the **energy** approach is used (the specific energy is constant at the outflow location):

$$\mathbf{dy/dx = [S_0 - S_f - (\alpha Q/(gA^2))\, q^*] / (1 - \alpha Q^2T/(gA^3)) \text{ with } q^* = dQ/dx \text{ (negative)}}$$

Side weir: $q = (2/3)\, C_d \sqrt{(2g)}\, (y - w)^{3/2}$ per unit length (w = weir height). In subcritical approach the depth rises along the weir; in supercritical approach it falls.

8.4 Applications of SVF

- Side-channel spillway for dam with long crest (energy-dissipation and passing floods).
- Side weirs in sewers (combined sewer overflows) and irrigation canals to divert excess flow.
- Bottom racks / Coanda screens in hydropower intakes (sediment exclusion).
- Roof and street gutters; drainage channels in highways and airports; Wash-water troughs.
- Perforated pipe manifolds (decreasing discharge).

✎ **Exam Question:** Derive the dynamic equation for spatially varied flow with increasing discharge. State the assumptions. (10 marks)

9. RAPIDLY VARIED FLOW (RVF) – APPLICATIONS

9.1 Characteristics

In RVF the curvature of streamlines is large; pressure is non-hydrostatic; boundary friction is small compared with the effect of rapid curvature. The analysis uses energy and momentum principles with empirical coefficients.

9.2 Hydraulic jump – design applications

- **Stilling basins (USBR types I–IV):** Type I for $Fr_1 < 1.7$ (simple apron); Type II for Fr > 4.5 and V > 15 m/s (chute blocks, dentated sill); Type III for V < 15 m/s (baffle blocks and end sill); Type IV for Fr = 2.5–4.5 (oscillating jump).
- **Jump on a sloped floor:** longer jump, position unstable.
- **Jump as an energy dissipator and mixer:** chemical dosing in water treatment, aeration.
- **Submerged jump** (drowned): when tailwater depth > y_2; used for sluice-gate outlets.

9.3 Sluice gate (under-gate flow)

Discharge per unit width: $\mathbf{q = C_d\, a\, \sqrt{(2g\, y_1)}}$, a = gate opening, y_1 = upstream depth; $C_d \approx 0.6$ for free flow. Contraction coefficient $C_c \approx 0.61$: vena contracta depth $y_{vc} = C_c\, a$. Downstream flow is supercritical; a jump forms if the tailwater is high enough.

9.4 Flow over weirs, transitions and drops

- **Free overfall (brink):** critical depth occurs about 3–4 y_c upstream of the brink; **brink depth $y_b \approx 0.715\, y_c$** (rectangular). A way to measure Q.
- **Channel transitions:** contraction/expansion; the design uses specific-energy or Froude-number approach; for supercritical flow, oblique shock waves (hence curved or straight-walled design).
- **Drop structures and chute blocks.**
- **Culvert flow:** inlet control versus outlet control; headwater depth.

9.5 Choking and critical-flow control

A constriction can drive the flow to critical depth; discharge becomes dependent only on the upstream head, so the structure acts as a control (flume, weir).

10. INTRODUCTION TO HEC-RAS

10.1 What is HEC-RAS?

HEC-RAS (Hydrologic Engineering Center's River Analysis System) is free software of the U.S. Army Corps of Engineers for 1-D (and 2-D) hydraulic modelling of rivers and channels. Capabilities:

- Steady-flow water-surface profile computation (standard step method).
- Unsteady flow simulation (Saint-Venant equations, implicit finite-difference).
- Sediment transport/movable boundary computations.
- Water-quality analysis; 2-D flood inundation mapping (RAS Mapper).

10.2 Data required

Data type	Details
Geometry	River centreline, cross-sections (station–elevation), bank stations, reach lengths (left, channel, right)
Roughness	Manning n for left overbank, channel, right overbank
Structures	Bridges, culverts, weirs, inline structures (dams)
Flow data	Peak discharges for each profile; unsteady hydrographs
Boundary conditions	Known WS, critical depth, normal depth (slope), rating curve
Loss coefficients	Contraction 0.1, expansion 0.3

10.3 Workflow for steady-flow analysis

- 1. Start a new project; set units (SI).
- 2. Enter geometric data (reach schematic → cross-sections → structures).
- 3. Enter steady-flow data and boundary conditions (subcritical: downstream; supercritical: upstream; mixed: both).
- 4. Run steady-flow analysis (choose flow regime).
- 5. View results: profile plot, cross-section plots, tables (WS elevation, velocity, Fr, energy grade, top width).
- 6. Check errors/warnings (balance error, critical depth, conveyance ratio), calibrate n.
- 7. Floodplain mapping and report.

10.4 Theoretical basis in brief

- **Energy equation** solved by the standard step method between consecutive cross-sections.
- **Momentum equation** is used where flow is rapidly varied: hydraulic jumps, bridges, stream junctions.
- **Conveyance** K is computed by subdividing cross-sections at n-value changes. Average friction slope: $S_f = ((Q_1 + Q_2)/(K_1 + K_2))^2$ (average conveyance), or other options.
- **Unsteady flow:** Saint-Venant continuity $\partial A/\partial t + \partial Q/\partial x = 0$ and momentum $\partial Q/\partial t + \partial(\beta Q^2/A)/\partial x + gA(\partial y/\partial x + S_f - S_0) = 0$ solved by a four-point implicit (Preissmann) scheme.

10.5 Quick revision – formulae

Topic	Formula
Manning	$Q = (1/n)AR^{2/3}S_0^{1/2}$
Froude number	$Fr = V/\sqrt{(gA/T)}$
Specific energy	$E = y + Q^2/2gA^2$
Critical depth (rectangle)	$y_c = (q^2/g)^{1/3}$, $E_{min} = 1.5y_c$
Specific force	$M = Q^2/gA + A\bar{y}$
Conjugate depths	$y_2/y_1 = ½(\sqrt{(1 + 8Fr_1^2)} - 1)$
GVF equation	$dy/dx = (S_0 - S_f)/(1 - Fr^2)$
Direct step	$\Delta x = \Delta E/(S_0 - S_f)$

23A01604b • OPEN CHANNEL FLOW

UNIT – 3

Unsteady Flows in Open Channels

Complete Study Notes • Concepts · Derivations · Formulae · Exam Questions

Syllabus Topics

- Equations of Motion (Saint-Venant equations)
- Uniformly Progressive Wave
- Rapidly Varied Unsteady Flow – Positive and Negative Surges
- Dam Break Problem

1. INTRODUCTION TO UNSTEADY OPEN-CHANNEL FLOW

1.1 What is unsteady flow?

Unsteady flow is flow in which depth and velocity change with time at a fixed location, i.e. $y = y(x, t)$ and $V = V(x, t)$. It is always non-uniform, because a change in time implies a change in space as the disturbance travels along the channel. Steady flow is only an idealisation; real channels carry flood waves, tidal bores, surges from gate operation, and dam-break waves.

1.2 Common examples

- **Flood waves** in rivers (slow, long waves caused by rainfall–runoff).
- **Surges** produced by sudden opening or closing of a gate, turbine load rejection in a power canal, or tidal bores in estuaries.
- **Dam-break waves** – extremely rapid and destructive.
- **Tides and seiches** in estuaries and lakes; **irrigation canal operation** (gate adjustments).

1.3 Classification of unsteady flow

Type	Feature	Example
Gradually varied unsteady	Long wave; vertical acceleration negligible; hydrostatic pressure	Flood routing
Rapidly varied unsteady	Abrupt change in depth; steep wave front; surface curvature important	Surges, bores, dam break

1.4 Wave celerity (small-amplitude long wave)

A small disturbance in still water of depth y moves with the **celerity**

$$\mathbf{c = \sqrt{g\,y}\ (rectangular\ channel)\ or\ c = \sqrt{g\,A/T} = \sqrt{g\,D}}$$

Relative to a moving stream of velocity V: absolute wave speeds are $\mathbf{V \pm c}$. If $Fr = V/c < 1$, a wave can travel upstream; if $Fr > 1$, all waves are swept downstream. This is why Fr is the key parameter in open-channel dynamics.

1.5 Methods of analysis

- **Saint-Venant equations** with characteristics or finite-difference methods (full dynamic wave).
- **Approximate routing** (kinematic wave, diffusion wave, Muskingum).
- **Surge theory** (momentum + continuity across the wave front).

2. EQUATIONS OF MOTION – THE SAINT-VENANT EQUATIONS

2.1 Assumptions

- Flow is one-dimensional: depth and velocity vary only along the channel axis x.
- Streamline curvature is small; vertical acceleration is negligible, therefore pressure is **hydrostatic**.
- Channel bed slope is small ($\cos\theta \approx 1$); channel is straight with fixed bed (no scour or deposition).
- Boundary friction is represented by steady-flow formulas (Manning or Chezy) using the instantaneous V and R.
- The fluid is incompressible and of constant density.

2.2 Continuity equation

Consider an elementary reach dx. Mass in – mass out = increase of storage. Let lateral inflow per unit length be q_L.

$$\partial Q/\partial x + \partial A/\partial t = q_L$$

In terms of V and y for a rectangular/general channel (T = dA/dy): $\partial y/\partial t + V\,\partial y/\partial x + D\,\partial V/\partial x = 0$ (for $q_L = 0$, D = A/T).

2.3 Momentum (dynamic) equation

Newton's second law applied to the CV between x and x + dx. Forces: hydrostatic pressure force difference, gravity component, boundary friction. Result:

$$(1/g)\,\partial V/\partial t + (V/g)\,\partial V/\partial x + \partial y/\partial x = S_0 - S_f$$

Every term has the dimension of slope (dimensionless). They are called:

	Name	Meaning		
	Local acceleration term	Change of velocity with time at a section		
	Convective acceleration term	Change of velocity along x due to non-uniformity		
	Pressure (hydrostatic) term	Water surface gradient		
	Gravity (bed slope) term	Weight component along channel		
	Friction slope term	Resistance, $S_f = n^2V$	V	$/R^{4/3}$

2.4 The conservation form

Dependent variables A and Q:

$$\partial A/\partial t + \partial Q/\partial x = 0$$

$$\partial Q/\partial t + \partial(\beta Q^2/A)/\partial x + g\,A\,\partial y/\partial x = g\,A\,(S_0 - S_f)$$

This form is better for numerical schemes (finite-volume, Preissmann) and for dealing with shocks and bores.

2.5 Special cases – hierarchy of wave models

By dropping terms from the momentum equation we get simplified models:

Model	Terms retained	Equation	Use
Kinematic wave	$S_0 = S_f$	Q = Q(A) (uniform flow relation)	Steep slopes, overland flow, no backwater
Diffusion wave	$\partial y/\partial x$, S_0, S_f	Neglect local and convective acceleration	Flood routing with backwater, mild slopes
Quasi-steady dynamic wave	Convective + pressure + S_0, S_f	Neglect local acceleration	Slowly changing flows
Full dynamic wave	All terms	Complete Saint-Venant	Dam break, tides, surges, steep fronts

★ **Key Point:** Always start from the full equations and then identify which terms are negligible for the problem. Kinematic wave cannot reproduce attenuation or backwater; diffusion wave can.

2.6 Solution of the Saint-Venant equations

- **Method of characteristics:** the two PDEs are converted into four ODEs along the characteristic directions dx/dt = V ± c.
- **Explicit finite-difference schemes** (Lax diffusive, MacCormack, Leap-frog) need Courant condition $\Delta t \leq \Delta x/(|V| + c)$.
- **Implicit schemes** (Preissmann four-point box scheme with weighting $\theta \approx 0.6$–1.0) – unconditionally stable, used in HEC-RAS, MIKE 11, SOBEK.
- **Finite-volume / Godunov schemes** with Riemann solvers – best for shocks, bores and dam-break.

2.7 Characteristic form (derivation outline)

Multiply the continuity equation (written with y) by a factor λ and add to the momentum equation, then choose λ such that the combination can be expressed as total derivatives. For a rectangular channel with $c = \sqrt{(gy)}$:

$$\mathbf{C^+: d(V + 2c)/dt = g(S_0 - S_f) \text{ along } dx/dt = V + c}$$

$$\mathbf{C^-: d(V - 2c)/dt = g(S_0 - S_f) \text{ along } dx/dt = V - c}$$

The quantities V ± 2c are called **Riemann invariants** (for a frictionless horizontal channel they stay constant along the characteristics). For $S_0 = S_f = 0$:

V + 2c = constant along C^+; V − 2c = constant along C^-.

These are the basis of the simple-wave solutions for dam-break and negative surges (Sections 4–5).

✎ **Exam Question:** Derive the continuity and dynamic equations for unsteady flow in open channels. List the assumptions. (10 marks)

3. UNIFORMLY PROGRESSIVE WAVE

3.1 Definition

A **uniformly progressive wave** (also called a **monoclinal wave** or **steady wave**) is a wave that moves along the channel with constant velocity V_w and without change of shape. Observed from a reference frame travelling with the wave, the flow appears steady, so the problem reduces to a steady non-uniform flow analysis.

Examples: flood wave in a long channel; a surge in which depth difference is constant; wave maintained by a constant inflow.

3.2 Transformation to steady flow

Let $\xi = x - V_w t$. For a wave of constant shape, $y = y(\xi)$, $Q = Q(\xi)$. Then $\partial/\partial t = -V_w \, d/d\xi$ and $\partial/\partial x = d/d\xi$. Substituting in the continuity equation $\partial Q/\partial x + \partial A/\partial t = 0$:

$dQ/d\xi - V_w \, dA/d\xi = 0 \rightarrow$ **$Q - V_w A =$ constant**.

That is, the discharge relative to the moving frame, $Q_r = (V - V_w)A$, is the same at every section.

3.3 Wave celerity (Kleitz–Seddon law)

Applying the constant-shape condition at two points on the wave (e.g. far upstream and far downstream where uniform flow exists with (Q_1, A_1) and (Q_2, A_2)):

$$\mathbf{V_w = (Q_2 - Q_1)/(A_2 - A_1) \rightarrow dQ/dA = (1/T)\, dQ/dy}$$

This is the **Kleitz–Seddon law**: the celerity of a flood wave equals the slope of the discharge rating curve divided by the top width. If the rating curve is the Manning relation $Q = (1/n)AR^{2/3}S_0^{1/2}$, then (wide channel, $R \approx y$)

$$\mathbf{V_w = (5/3)\ V\ (wide\ rectangular)\ ;\ (13/9)\ V\ (wide\ parabolic)\ ;\ (4/3)\ V\ (triangular)\ [Manning]}$$

With **Chezy** friction a wide rectangular channel gives $V_w = (3/2)V$. So a flood wave moves faster than the water particles – the wave transmits "information" or kinematic effect and the water just passes through.

★ **Key Point:** Relationship between flow velocity and kinematic wave speed: $c_k = dQ/dA = \beta_k V$, where β_k = 1.3–1.7 (Manning). Typical flood waves travel at 1–3 m/s in lowland rivers.

3.4 Dynamic uniformly progressive wave

Using the dynamic equation with Chezy friction for a wave with $\partial y/\partial \xi \neq 0$, the water surface profile of the wave is given by the integration of

$$\mathbf{dy/d\xi = (S_0 - S_f) \,/\, (1 - (V - V_w)^2/(g\, y))}$$

This looks like the GVF equation with the velocity replaced by the **relative velocity**. A smooth monoclinal wave exists only if the relative Froude number $(V - V_w)/\sqrt{(gy)}$ at the wave is < 1 at the upstream end of the wave; if V_w is such that Fr_r exceeds unity, the front steepens into a bore (hydraulic jump that moves).

3.5 Condition for stability (Vedernikov number)

Roll waves (trains of surges) develop on steep, smooth channels (chutes, spillways) when the Froude number exceeds a limit given by the **Vedernikov–Powell criterion**: for a wide channel, Fr > 2 with Chezy friction and Fr > 1.5 with Manning friction. Roll waves are periodic uniformly progressive waves with hydraulic jumps; they can cause overtopping of chute walls.

3.6 Applications

- Flood routing by kinematic wave in steep catchments and overland flow.
- Estimating flood travel time $T = L/V_w$ in a river reach.
- Irrigation-canal operation: gate-change waves travel with celerity $V \pm c$; control by downstream-gate operation.

✎ **Exam Question:** A flood wave moves in a wide rectangular channel with n = 0.03, S_0 = 0.0004. Normal depth is 2.0 m. Find wave speed. *Ans: $V = (1/n)\, y^{2/3}\, S^{1/2} = (1/0.03)(1.587)(0.02) = 1.058$ m/s; $V_w = 5/3 \times 1.058 = 1.76$ m/s.*

4. RAPIDLY VARIED UNSTEADY FLOW – SURGES

4.1 Definition

A **surge (or wave) of finite height** is a moving hydraulic jump produced when the flow is suddenly changed at a section (gate closing/opening, turbine rejection, tidal inflow). It is a rapidly varied unsteady flow because the water surface rises or falls abruptly at the wave front; hydrostatic assumption is not valid in the front.

- **Positive surge:** the depth increases across the wave ($y_2 > y_1$).
- **Negative surge:** the depth decreases across the wave ($y_2 < y_1$).

Further, according to the direction of travel of the wave front: **upstream-moving** (due to downstream gate closing or inflow reduction/increase upstream) and **downstream-moving** (due to gate opening upstream, or sudden increases in inflow at upstream end).

Cause	Surge type	Direction of travel
Sudden closure of downstream gate	Positive surge	Upstream
Sudden partial closure of upstream gate	Negative surge	Downstream
Sudden opening of upstream gate / increased inflow	Positive surge	Downstream
Sudden opening of downstream gate	Negative surge	Upstream

4.2 Positive surge – basic equations

Consider a surge moving upstream into a channel with steady flow V_1, depth y_1. Behind the surge the flow has V_2 and depth y_2. The surge wave speed (celerity) relative to the bank is V_w. A moving observer rides with the surge so the flow looks steady.

Relative velocities: upstream of surge: $(V_1 + V_w)$, downstream: $(V_2 + V_w)$.

Continuity: $y_1(V_1 + V_w) = y_2(V_2 + V_w)$ (1)

Momentum (per unit width):

$(\gamma/2)(y_1^2 - y_2^2) = \rho\, y_1(V_1 + V_w)[(V_2 + V_w) - (V_1 + V_w)]$ (2)

Eliminating V_2 between (1) and (2):

$$\mathbf{V_w = -V_1 + \sqrt{\left(g\, y_2\, (y_1 + y_2) / (2\, y_1) \right)}}$$

Equivalent forms: $\mathbf{V_w + V_1 = \sqrt{(g\, y_1)}\ \sqrt{[\ ½\ (y_2/y_1)(1 + y_2/y_1)\ ]}}$. For a still-water upstream ($V_1 = 0$) the surge speed is

$$\mathbf{V_w = \sqrt{\left(g\, y_2 (y_1 + y_2)/(2y_1) \right)} \geq \sqrt{(g\, y_1)}}$$

which equals the small-wave celerity when $y_2 \to y_1$. So the larger the surge height, the faster the surge. From continuity (1) the velocity behind the surge is

$$\mathbf{V_2 = y_1(V_1 + V_w)/y_2 - V_w}$$

For still water ahead ($V_1 = 0$) this gives $V_2 = -V_w(y_2 - y_1)/y_2$: the negative sign shows that the water behind a tidal bore moves in the same direction as the bore (upstream).

4.3 Energy loss across the surge

$$h_L = (y_2 - y_1)^3 / (4\, y_1\, y_2)$$

Same expression as the stationary hydraulic jump, since in the moving frame it is a steady jump. The Froude number of the surge: $Fr_s = (V_1 + V_w)/\sqrt{(g\, y_1)}$; if $Fr_s < 1.7$ the front is **undular** (a train of waves following – "undular bore"), $Fr_s > 1.7$ turbulent with white-water roller (e.g., Severn bore, Qiantang bore).

4.4 Worked example – positive surge

A canal flows at $V_1 = 1.0$ m/s with $y_1 = 2.0$ m. A downstream gate is suddenly partly closed and the depth behind the surge becomes $y_2 = 2.4$ m. Find the surge speed and the velocity behind it.

- $V_w = -1.0 + \sqrt{(9.81 \times 2.4 \times 4.4/(2 \times 2.0))} = -1.0 + \sqrt{25.9} = -1.0 + 5.09 =$ **4.09 m/s** (moving upstream relative to the bank).
- Continuity: $V_2 = y_1(V_1 + V_w)/y_2 - V_w = 2.0 \times 5.09/2.4 - 4.09 = 4.24 - 4.09 =$ **0.15 m/s** (downstream). The closure has cut the velocity from 1.0 to 0.15 m/s.
- Energy loss: $h_L = (0.4)^3/(4 \times 2.0 \times 2.4) = 0.0033$ m ($Fr_s = 5.09/4.43 = 1.15$, so the front is undular).

★ **Key Point:** Always draw a diagram with arrows and define signs before applying the continuity and momentum equations. Most mistakes in surge problems are sign errors.

5. NEGATIVE SURGES AND SIMPLE WAVES

5.1 Negative surge

In a negative surge the depth falls across the wave; unlike the positive surge, **there is no stable abrupt front** – energy would have to increase across a "negative jump", which is impossible. Hence a negative surge is a **smooth, gradually flattening wave**, i.e. a depression wave in which each depth travels with its own celerity (a rarefaction fan). The wave flattens as it travels because the deeper water at its leading edge has a larger celerity than the shallower water behind it.

5.2 Analysis using characteristics

Take a frictionless horizontal rectangular channel, initial V_1, y_1 and a flow reduction or lowered depth created at one end. The disturbance is a **simple wave**: all characteristics of one family (here C^+) start in the region of uniform flow, so the Riemann invariant carried by that family has the same value everywhere in the wave.

For a negative wave travelling upstream (into region 1 with y_1, V_1), the invariant along C^+ is constant:

$$\mathbf{V + 2\sqrt{(g\,y)} = V_1 + 2\sqrt{(g\,y_1)} = constant}$$

So the velocity at any depth y in the wave is $\mathbf{V = V_1 + 2(\sqrt{(g\,y_1)} - \sqrt{(g\,y)})}$ ($y < y_1$, so velocity rises as depth falls). The C^- characteristics are straight lines with slope $\mathbf{dx/dt = V - \sqrt{(gy)} = V_1 + 2\sqrt{(g\,y_1)} - 3\sqrt{(gy)}}$. The leading edge (depth y_1) travels at $V_1 - \sqrt{(g\,y_1)}$ (upstream); a lower depth travels more slowly, so the wave spreads out as it advances.

5.3 Comparison

Feature	Positive surge	Negative surge
Depth change	Rise ($y_2 > y_1$)	Fall ($y_2 < y_1$)
Shape of front	Steep, abrupt (moving jump)	Gradual, flattening (spreads)
Energy	Lost across the front ($h_L > 0$)	Conserved (isentropic)
Method	Continuity + momentum across jump	Method of characteristics (Riemann invariants)
Celerity	Same everywhere at the front	Varies with depth, $V \pm \sqrt{(gy)}$
Example	Tidal bore; gate closing	Gate opening downstream; sudden pump shutoff upstream

5.4 Applications and engineering significance

- Design of **freeboard** and canal lining: positive surges following turbine rejection in power canals may overtop banks.
- **Power canals/penstock forebays:** upstream surge, negative surge on load demand increase.
- **Irrigation canal gates:** operation rules to limit surge height (≤ 10–20 % of normal depth).
- **Tidal bores:** navigation hazard; used for surfing; tidal power; sediment resuspension.
- **Pumping station start and stop;** emergency closure gates.

✎ **Exam Question:** Distinguish between positive and negative surges. Derive the expression for celerity of a positive surge in a rectangular channel. (10 marks)

6. DAM BREAK PROBLEM

6.1 Introduction

A **dam break** is a catastrophic failure of a dam which suddenly releases the stored water as a flood wave, travelling downstream. Examples: Teton Dam (1976), Banqiao (1975), Machhu-II (Morbi, India, 1979), Malpasset (France, 1959), St Francis (USA, 1928). Dam-break analysis serves **emergency action plans (EAPs)**, flood inundation mapping and risk assessment.

Failure modes: overtopping, piping/internal erosion, foundation failure, earthquake, landslide-generated waves.

Types: **instantaneous failure** (concrete arch or gravity dams – complete, sudden) and **progressive failure** (earth/rockfill – a breach grows over minutes to hours).

6.2 Ritter's solution – instantaneous total collapse on a dry bed

Assumptions: horizontal, frictionless, wide rectangular channel; dam removed instantly at t = 0; initial still water depth y_0 upstream; dry bed downstream. Hydrostatic pressure; 1-D flow.

Initial: V = 0, y = y_0 for x < 0; y = 0 for x > 0 (dam at x = 0). Reservoir celerity $c_0 = \sqrt{(g\, y_0)}$.

Characteristics: the **negative wave** moves upstream into the reservoir with speed $-c_0$ (i.e. the leading edge of the disturbance is at $x = -c_0\, t$). Since the region upstream is initially at rest, the Riemann invariant V + 2c along C^+ carries the value $2c_0$:

$$\mathbf{V + 2c = 2c_0 \rightarrow V = 2(c_0 - c)}$$

and the C^- characteristics are straight lines: $dx/dt = V - c = 2c_0 - 3c$, so $x/t = 2c_0 - 3c$. Solve for c and V:

$$\mathbf{c = (2c_0 - x/t)/3 \rightarrow y(x,t) = (1/9g)(2c_0 - x/t)^2}$$

$$\mathbf{V(x,t) = (2/3)(c_0 + x/t)}$$

valid for $\mathbf{-c_0 t \le x \le 2c_0 t}$. Wave front on the dry bed travels at $\mathbf{2c_0 = 2\sqrt{(g\, y_0)}}$; the water surface profile is a **parabola**.

Location	Depth	Velocity
$x = -c_0 t$ (upstream limit of disturbance)	y_0	0
x = 0 (dam site)	$\mathbf{4y_0/9} = 0.444 y_0$	$\mathbf{(2/3)c_0} = (2/3)\sqrt{(g y_0)}$
$x = 2c_0 t$ (front)	0	$2c_0$

Discharge at the dam site: $\mathbf{q = y\, V = (8/27)\, y_0\, \sqrt{(g\, y_0)}}$ per unit width, and it is **constant in time** (until the negative wave reaches the far end of the reservoir). Fr at x = 0 is exactly 1 – critical flow at the dam site.

★ **Key Point:** Ritter's solution overestimates the speed of the front since friction is neglected; with friction, the front is blunt, thick and travels slower than $2c_0$ (Dressler 1952).

6.3 Worked example

A dam holding y_0 = 40 m of water fails suddenly; the tailwater is dry. $c_0 = \sqrt{(9.81 \times 40)} = 19.8$ m/s. Depth at dam site = 17.8 m; velocity = 13.2 m/s; discharge per metre width = 235 m^2/s; front speed = 39.6 m/s (142 km/h, theoretical). After 5 minutes the front would be 11.9 km downstream (without friction).

7. DAM BREAK – WET BED (STOKER'S SOLUTION) AND PARTIAL BREACH

7.1 Dam break on a wet bed (tailwater depth y_1)

When there is initial water of depth y_1 ($< y_0$) downstream, a **positive surge** (bore) moves downstream with speed V_w and a **negative wave** moves upstream into the reservoir. Between them is a zone of constant depth y_m and velocity V_m.

Negative wave (Riemann invariant from reservoir): $V_m = 2(c_0 - c_m)$, $c_m = \sqrt{(g\, y_m)}$.

Positive surge (jump conditions): $V_w = V_m\, y_m/(y_m - y_1)$ and $V_w^2 = g\, y_m(y_m + y_1)/(2y_1)$.

Combining these gives a single non-linear equation for y_m (solve by Newton–Raphson):

$$\mathbf{2(c_0 - c_m) = (y_m - y_1)\,\sqrt{[\, g\,(y_m + y_1)/(2\, y_m\, y_1)\,]}}$$

Example: $y_0 = 10$ m, $y_1 = 1$ m. Here $c_0 = 9.9$ m/s; iteration gives $y_m \approx 3.96$ m, $V_m \approx 7.35$ m/s and bore speed $V_w \approx 9.8$ m/s. Ratio $y_1/y_0 \to 0$ gives the dry-bed limit; ratio $\to 1$ gives a weak wave.

7.2 Effect of friction and channel slope

Real dam-break waves attenuate (peak discharge decreases and wave flattens) as they move downstream due to resistance, storage in the valley (floodplain), channel widening and loss of volume. For analysing real valleys:

- Use the full Saint-Venant equations with implicit or Godunov schemes.
- Use dimensionless parameters: $Q_* = Q/Q_{max}$, distance $X_* = x/L$.
- Use empirical formulas for peak outflow from breach (Froehlich 1995): $Q_p = 0.607\, V_{res}^{0.295}\, h^{1.24}$ (SI), V_{res} = reservoir volume (m^3), h = height of water above the breach bottom (m).

7.3 Peak flow estimation

- **Instantaneous failure** of a rectangular section: $Q_{max} = (8/27)\, b\, \sqrt{g}\, y_0^{3/2}$ (Ritter).
- **Linear breach with time:** the breach width grows linearly; the peak flow from the broad-crested weir equation $Q = C\, b\, y^{3/2}$.
- **Peak attenuation:** the peak falls as the wave moves down the valley because of valley storage and friction; it is estimated by storage routing (e.g. Wetmore–Fread) or numerical models.

7.4 Dam-break analysis workflow

- 1. Collect reservoir data: volume–elevation curve, dam geometry, breach scenario (sunny-day, flood overtopping).
- 2. Estimate breach parameters: width, side slopes, formation time (empirical equations – MacDonald & Langridge-Monopolis, Froehlich, Xu & Zhang).
- 3. Compute outflow hydrograph through the breach (weir-flow, reservoir routing).
- 4. Route through downstream valley (HEC-RAS 1D/2D, MIKE 11, DAMBRK, BREACH, FLO-2D).
- 5. Map flood extent, depth, velocity, arrival time and hazard classification.
- 6. Prepare **Emergency Action Plan**: inundation maps, warning, evacuation, communication.

7.5 Dam-break and surge – key comparisons

Aspect	Dry-bed (Ritter)	Wet-bed (Stoker)
Front type	Smooth, tip with zero depth	Steep bore (jump) of finite height
Front speed	$2c_0$	$V_w < 2c_0$, from jump equation
Depth at dam site	$4y_0/9$ (constant)	$4y_0/9$ if $y_m < 4y_0/9$, otherwise y_m
Method	Characteristics (simple wave)	Characteristics + jump conditions

✎ **Exam Question:** Derive Ritter's solution for the instantaneous dam-break problem on a dry horizontal bed. Show that the depth at the dam site is 4/9 y_0. (10 marks)

8. NUMERICAL METHODS AND QUICK REVISION

8.1 Numerical solution of the Saint-Venant equations

Explicit schemes: simple, but stable only if Courant number Cr = (|V| + c)Δt/Δx ≤ 1. Lax diffusive scheme, MacCormack predictor–corrector scheme (second-order accurate, good for shocks with artificial viscosity).

Implicit schemes: Preissmann box scheme; weighted averages of variables at n and n+1 time levels give a banded (block-tridiagonal) system solved by the double-sweep method. Stable for θ ≥ 0.5; Δt can be large (but accuracy reduces).

Finite-volume / Godunov: solves Riemann problem at each cell interface (Roe, HLL, HLLC solvers); handles discontinuities and wet–dry fronts. Used in modern dam-break models (HEC-RAS 2D, TELEMAC, LISFLOOD).

8.2 Boundary and initial conditions

- **Initial:** steady-flow profile (backwater calculation) for all x.
- **Upstream boundary:** inflow hydrograph Q(t) (flood) or stage hydrograph.
- **Downstream boundary:** rating curve Q(y), stage hydrograph (tide) or normal depth.
- **Internal:** gates, weirs, bridges, culverts – given by head–discharge relations.

8.3 Flood routing methods in brief

Method	Equation	Parameters
Kinematic wave	$\partial A/\partial t + c_k\, \partial A/\partial x = 0$	$c_k = dQ/dA$
Muskingum	$S = K[XI + (1-X)O]$	K (travel time), X (0–0.5)
Diffusion wave	$\partial Q/\partial t + c\, \partial Q/\partial x = D\, \partial^2 Q/\partial x^2$	Celerity and diffusivity $D = Q/(2TS_0)$
Dynamic wave	Full Saint-Venant	Requires numerical scheme

8.4 Formulae sheet

	Formula		
	$c = \sqrt{(gy)}$		
	$\partial A/\partial t + \partial Q/\partial x = 0$		
	$S_f = S_0 - \partial y/\partial x - (V/g)\partial V/\partial x - (1/g)\partial V/\partial t$		
	$V_w = dQ/dA = (1/T)dQ/dy$		
	$V_w = -V_1 + \sqrt{(gy_2(y_1 + y_2)/2y_1)}$		
	$h_L = (y_2 - y_1)^3/(4y_1y_2)$		
	$y_{dam} = 4y_0/9$; front speed $= 2\sqrt{(gy_0)}$		
	Δt ≤ Δx/(	V	+ c)

8.5 Important short questions

- What are the assumptions of the Saint-Venant equations?
- Define uniformly progressive wave. State the Kleitz–Seddon law.
- Why does a negative surge not form an abrupt front?
- What are Riemann invariants?
- State the Courant–Friedrichs–Lewy (CFL) condition.
- Differentiate kinematic, diffusion and dynamic wave models.

- List causes of dam failure and two methods of analysis.

23A01604b • OPEN CHANNEL FLOW

UNIT – 4

Sediment Transport

Complete Study Notes • Concepts · Derivations · Formulae · Exam Questions

Syllabus Topics

- Sediment Properties
- Inception of Sediment Motion
- Bed Forms
- Bed Load, Suspended Load and Total Sediment Transport
- Design of Stable Channels and Regime Channels
- Reservoir Sedimentation and Trap Efficiency

1. INTRODUCTION TO SEDIMENT TRANSPORT

1.1 Why study sediment?

Rivers and canals carry water and sediment together. Water flowing over an erodible boundary can scour the bed and banks, carry the material, and deposit it elsewhere. Sediment transport governs river morphology, silting of canals and reservoirs, scour around bridge piers, stability of channels and the life of dams.

1.2 Sources and classification of sediment

- **By origin:** erosion from catchment (sheet, rill, gully), bank erosion, bed erosion; wash load comes mostly from the catchment.
- **By mode of transport:** (a) **bed load** – rolling, sliding, saltating near the bed; (b) **suspended load** – supported by turbulence; (c) **wash load** – very fine particles (< 0.0625 mm) that stay in suspension and are not found in the bed material.
- **By source: bed-material load** = bed load + the suspended part of bed material; it depends on the flow and can be predicted. **Wash load** depends on supply from the catchment, not on the transport capacity.

★ **Key Point:** Total load = bed load + suspended load = bed-material load + wash load. Typically, bed load is 5–25 % of the total load in sand-bed rivers.

1.3 Types of sediment-transport problems

- Alluvial channels (bed and banks are made of the transported material): design of stable canals, regime channels.
- Aggradation and degradation of the bed (silting upstream of weirs, scouring downstream of dams).
- Local scour (bridge piers, abutments, aprons).
- Reservoir and canal sedimentation.

2. SEDIMENT PROPERTIES

2.1 Particle size

Size is the most important property. Defined by the **sieve diameter** (sand and gravel) or **sedimentation diameter** (silt and clay). Grain size classes (modified Wentworth scale):

Class	Size range (mm)	Remark
Boulder	> 256	Moves only in extreme floods
Cobble	64 – 256	Mountain streams
Gravel	2 – 64	Moves as bed load
Sand	0.0625 – 2	Bed + suspended load
Silt	0.004 – 0.0625	Mainly suspended
Clay	< 0.004	Cohesive; wash load

Characteristic sizes: d_{50} (median; 50 % finer by weight), d_{35}, d_{65}, d_{84}, d_{90}. **Geometric mean** $d_g = \sqrt{(d_{84}\, d_{16})}$; **gradation coefficient** $\sigma_g = \sqrt{(d_{84}/d_{16})}$ ($\sigma_g < 1.3$: uniform; > 3: poorly sorted). Sieve analysis gives the **grain-size distribution curve** (percent finer vs log d).

2.2 Shape and density

- **Shape factor** SF = c/√(ab), where a, b, c are the longest, intermediate and shortest axes (natural sand ≈ 0.7). **Sphericity, roundness.**
- **Specific gravity** $s = \rho_s/\rho \approx 2.65$ for quartz; submerged specific weight $\gamma' = (s - 1)\gamma = 16.2$ kN/m^3.
- **Porosity** $p = V_{void}/V_{total} \approx 0.35$–$0.45$ for sand; **dry bulk density** $= (1 - p)\,\rho_s$; dry density of deposited sediment (reservoir): sand 1500–1700 kg/m^3, silt 1000–1400, clay 500–900.
- **Angle of repose φ:** 30°–40° for sand; 35°–45° for gravel; increases with grain size and angularity.

2.3 Fall (settling) velocity

Velocity at which a particle settles in still water, when drag = submerged weight. Drag coefficient C_D depends on particle Reynolds number $Re_p = w\,d/\nu$.

Stokes ($d < 0.1$ mm, $Re_p < 1$): $w = (s - 1)\, g\, d^2 / (18\, \nu)$

Rubey (general, natural sand): $w = F\sqrt{[(s - 1)\, g\, d]}$, $F = \sqrt{(2/3 + 36\nu^2/((s - 1)g\, d^3))} - \sqrt{(36\nu^2/((s - 1)g\, d^3))}$

Large grains ($Re_p > 1000$): $w = 1.1\sqrt{[(s - 1)\, g\, d]}$. Typical values for quartz in water at 20 °C: d = 0.1 mm → 0.009 m/s; 0.5 mm → 0.07 m/s; 1 mm → 0.12 m/s; 10 mm → 0.44 m/s. Fall velocity is reduced by high concentration (hindered settling), turbulence, and increased by flocculation of fines.

2.4 Dimensionless parameters

Particle Reynolds (shear) number: $Re_* = u_*\, d / \nu$

Dimensionless grain diameter: $D_* = d\,[(s - 1)\, g / \nu^2]^{1/3}$

Shields (mobility) parameter: $\tau_* = \tau_0 / [(\gamma_s - \gamma)\, d]$

Here **shear velocity** $u_* = \sqrt{(\tau_0/\rho)} = \sqrt{(g\ R\ S_0)}$ for uniform flow. $Re_* < 5$: hydraulically smooth; $Re_* > 70$: fully rough bed.

3. INCEPTION OF SEDIMENT MOTION

3.1 Forces on a bed particle

A particle resting on the bed experiences the **drag force** F_D and **lift force** F_L from the flow, and its **submerged weight** W' resisting movement through friction at contact points (tan φ). Motion begins when the moment of the driving forces about a contact point exceeds the moment of the resisting forces. Because turbulence makes the instantaneous forces fluctuate, the "critical" condition is statistical.

3.2 Critical shear stress – Shields criterion

Dimensional analysis shows that at incipient motion

$$\tau_{*c} = \tau_c / [(\gamma_s - \gamma)\, d] = f(Re_*)$$

Shields diagram is the plot of τ_{*c} against Re_*. Its main features:

Region	Re_*	τ_{*c}	Remark
Hydraulically smooth	< 5	≈ 0.1 and falling as Re_* rises	Particle buried in viscous sublayer
Transition	5 – 70	min ≈ 0.03 at Re_* ≈ 10	Lowest τ_{*c}
Fully rough	> 70 (> 400)	constant ≈ 0.045 – 0.06	Typical gravel and coarse sand

So for coarse particles: $\tau_c = 0.045\,(\gamma_s - \gamma)\, d$. For quartz in water: $\tau_c \approx 0.045 \times 16\,200 \times d = 730\, d$ (Pa, d in m) → $\tau_c \approx 0.73\, d$(mm) Pa. For d = 1 mm, $\tau_c \approx 0.5$–0.8 Pa.

Critical velocity: from $u_{*c} = \sqrt{(\tau_c/\rho)}$ and a logarithmic velocity profile (or Manning) the mean critical velocity can be derived; empirical relations (Neill, Garde) have the form $V_c = K\, y^{1/6}\, d^{1/3}$ with K ≈ 6 (SI units).

3.3 Procedure to check incipient motion (design)

- 1. Compute $\tau_0 = \gamma\, R\, S$ (or $\gamma\, y\, S$ for wide channel).
- 2. Compute $Re_* = u_* d/\nu$ and read τ_{*c} from Shields diagram (or use 0.045).
- 3. Compute $\tau_c = \tau_{*c}(\gamma_s - \gamma)d$.
- 4. If $\tau_0 > \tau_c$ the bed moves; otherwise stable. Ratio τ_0/τ_c = **excess shear**.

3.4 Effect of bed slope and side slope

On a bank of slope angle θ (θ < φ) the critical shear stress is reduced by factor K:

$$K = \tau_{c,side} / \tau_{c,bed} = \sqrt{(1 - \sin^2\theta / \sin^2\varphi)} = \cos\theta\, \sqrt{(1 - \tan^2\theta/\tan^2\varphi)}$$

Example: φ = 35°, side slope 1.5H : 1V (θ = 33.7°): $K = \sqrt{(1 - (0.555/0.574)^2)} = 0.25$, i.e. bank particles are much less stable.

3.5 Other criteria

- **White (1940):** turbulence-based, includes hiding effect.
- **Yalin:** τ_{*c} as function of D_*.
- **Shields formula for gradation:** use d_{50} for uniform sand; for mixtures, finer fraction hide behind coarse ones (**hiding factor**), and coarse fractions are more exposed. Armouring occurs when finer fractions are winnowed away, leaving a coarse protective layer (e.g., d_{90} layer).
- **Permissible (Lane/Fortier-Scobey) velocity** used in practice for canals (see Section 8).

✎ **Exam Question:** Explain Shields' criterion for initiation of sediment motion. Determine whether a sand bed with d = 2 mm will move in a wide channel of depth 1.2 m and slope 0.0005. *Ans:* $\tau_0 = 9810 \times 1.2 \times 0.0005 = 5.9$ *Pa.* $\tau_c = 0.045 \times 16\,200 \times 0.002 = 1.46$ *Pa.* $\tau_0 > \tau_c$ *(4× larger), so the bed material moves.*

4. BED FORMS

4.1 Why bed forms form

In an alluvial channel with sand beds, the bed does not remain flat once sediment begins moving. The flow interacts with the bed and creates regular patterns called **bed forms**. They change flow resistance (Manning n and friction factor), sediment transport rate and water level, so they matter in design and in flood stage prediction.

4.2 Types of bed forms (flow regime concept – Simons & Richardson)

Regime	Bed form	Fr	Water surface relation	Resistance / n
Lower regime	Ripples (d < 0.6 mm), ripples on dunes, dunes	< 1 (Fr 0.1–0.6)	Out of phase with bed	High: n ≈ 0.018–0.035
Transition	Washed-out dunes / undulating bed	≈ 0.6–1	Unstable	Falls sharply with V
Upper regime	Plane bed (flat)	0.8–1.2	—	Low: n ≈ 0.012–0.016
Upper regime	Antidunes (standing waves)	> 1	In phase with bed	Low
Upper regime	Chutes and pools	> 1.5	Rapids and jumps	Moderate–high

- **Ripples:** small (wavelength < 0.6 m, height < 0.06 m) triangular forms with a gentle upstream slope and steep lee face; they appear only for fine sand (d < 0.6 mm); they migrate downstream.
- **Dunes:** larger (wavelength 0.6–1000 m, height 0.06–10+ m), out-of-phase with the water surface. Sediment moves up the stoss side, avalanches down the lee, and so the dune moves downstream; flow separation behind the crest produces form drag. Height may be up to y/6; wavelength 5y.
- **Flat (plane) bed:** at higher velocity, dunes wash out; skin friction only; resistance is low, transport is high.
- **Antidunes:** surface and bed waves in phase; waves may move upstream, downstream or stand still; breaking waves occur.

4.3 Resistance to flow – skin and form friction

Einstein & Barbarossa: total resistance R = R' + R'' (R' = grain/skin; R'' = form). Only the grain-resistance part contributes to sediment transport. Strickler: $n' = d_{50}^{1/6}/21$ (d in m) or $n' = d_{90}^{1/6}/26$.

★ **Key Point:** In the lower regime the water surface level rises (resistance increases) when the discharge increases only slightly; in the transition the stage may drop with increasing discharge (discontinuous rating curve) because dunes wash out.

4.4 Prediction of bed-form type

Use diagrams: Simons–Richardson (stream power vs d_{50}), Liu, Van Rijn (D_* vs transport stage parameter $T = (\tau'_b - \tau_c)/\tau_c$: roughly ripples for T < 3, dunes for 3 < T < 15, plane bed for T > 15), Engelund–Hansen.

5. BED LOAD TRANSPORT

5.1 Definition

Bed load is the sediment that moves in continuous or intermittent contact with the bed by rolling, sliding and hopping (saltation). It is expressed as q_b (mass or weight or volume per unit width per unit time). The bed-load layer is only a few grain diameters thick.

5.2 Du Boys formula (1879)

Sediment moves as layers of thickness proportional to the excess shear stress:

$q_b = \psi\, \tau_0\, (\tau_0 - \tau_c)$ where ψ is the Du Boys characteristic coefficient (depends on grain size, found by Straub from flume data) and τ_c comes from Shields. It is the oldest bed-load formula and is mainly of historical value now.

5.3 Meyer-Peter and Müller (MPM, 1948)

The most widely used formula for gravel and coarse sand (d from 0.4 to 30 mm):

$$\Phi = q_b / \sqrt{((s-1)\, g\, d^3)} = 8\, (\tau_*' - 0.047)^{3/2}$$

where Φ is the **dimensionless bed-load intensity** (q_b is the volumetric rate per unit width) and $\tau_*' = \tau_0'/((\gamma_s - \gamma)\, d)$ is the Shields parameter based on **grain** shear stress $\tau_0' = \gamma\, R'\, S$ (skin friction only). In terms of weight: $g_b = \gamma_s\, q_b$.

Worked example: d = 5 mm gravel, s = 2.65, wide channel depth 1.0 m, S = 0.002, no bedforms ($\tau_0' = \tau_0$). $\tau_0 = 9810 \times 1.0 \times 0.002 = 19.6$ Pa; $\tau_* = 19.6/(16\,200 \times 0.005) = 0.242$. $\Phi = 8(0.242 - 0.047)^{1.5} = 8 \times 0.0861 = 0.689$. $\sqrt{((s-1)g\, d^3)} = \sqrt{(1.65 \times 9.81 \times 1.25\times10^{-7})} = 1.42 \times 10^{-3}$ m^2/s. $q_b = 0.689 \times 1.42\times10^{-3} = 9.8\times10^{-4}$ m^3/s/m ≈ 2.6 kg/s/m (with $\rho_s = 2650$).

5.4 Einstein bed-load function (1950)

A **probabilistic** approach: the probability p that the instantaneous lift force on a grain exceeds its submerged weight is related to the rate of erosion, which is balanced by deposition. The dimensionless transport intensity Φ_* and flow intensity $\Psi = (s - 1)d/(R'S)$ are related by a single curve:

$$1 - (1/\sqrt{\pi}) \int e^{-t^2}\, dt = A_* \Phi_* / (1 + A_* \Phi_*) \quad (\text{limits} \pm (B_* \Psi - 1/\eta_0))$$

with $A_* = 43.5$, $B_* = 0.143$, $\eta_0 = 0.5$. Valid for sand and gravel; includes hiding and lift correction.

5.5 Other formulae

Formula	Form	Remark
Bagnold (1956)	$q_b \propto \omega$ (stream power)	Power approach; efficiency $e_b \approx$ 0.1–0.15
Einstein–Brown	$\Phi = 40\, \tau_*^3$	For $\tau_* > 0.18$
Van Rijn (1984)	$q_b = 0.053\, \sqrt{((s-1)g)}\, d_{50}^{1.5}\, D_*^{-0.3}\, T^{2.1}$	Sand 0.2–2 mm, T = transport stage
Parker, Wilcock–Crowe	Fractional transport	Gravel-bed rivers
Wong–Parker (2006)	$\Phi = 4.93(\tau_* - 0.0495)^{1.6}$	Improved MPM

6. SUSPENDED LOAD TRANSPORT

6.1 Mechanism

When the turbulence intensity (vertical fluctuations ≈ u_*) exceeds the fall velocity w, particles remain in suspension. Upward turbulent diffusion balances downward settling at equilibrium:

$$w\,C + \varepsilon_s\,dC/dy = 0$$

C = volumetric (or mass) concentration at height y above the bed; ε_s = sediment diffusion coefficient.

6.2 Rouse equation (vertical concentration profile)

Assume $\varepsilon_s = \beta\,\varepsilon_m$ with parabolic eddy viscosity $\varepsilon_m = \kappa\,u_*\,y\,(1 - y/h)$ ($\kappa = 0.4$, $\beta \approx 1$). Then integration gives

$$C/C_a = [\,((h - y)/y) \times (a/(h - a))\,]^z,\quad z = w / (\beta\,\kappa\,u_*)$$

C_a = reference concentration at height a above the bed (usually a = 0.05 h or 2d); h = flow depth; z = **Rouse number**.

Rouse number z	Mode of transport
> 2.5	Bed load
1.2 – 2.5	Suspended load (50 % of the flow depth)
0.8 – 1.2	Suspended load (100 % depth)
< 0.8	Wash load

★ **Key Point:** Larger z (heavy particles, low shear) gives a steep concentration profile concentrated near the bed; small z (fine particles, strong turbulence) gives uniform concentration over depth.

6.3 Suspended load rate

$$q_s = \int_a^h C(y)\,u(y)\,dy$$

Using the log-law velocity $u(y) = (u_*/\kappa)\,\ln(30y/k_s)$ and the Rouse profile, Einstein derived $q_s = 11.6\,u_*\,C_a\,a\,[P_E\,I_1 + I_2]$, where I_1, I_2 are Einstein integrals (tabulated) in terms of z and a/h, and $P_E = 2.303\,\log(30.2\,h/\Delta)$.

6.4 Measurement and analysis

- **Point-integrating or depth-integrating samplers** (US D-74, DH-48) collect water–sediment mixture. Concentration in mg/L (ppm). **Sediment discharge:** Q_s (t/day) = 0.0864 × C (mg/L) × Q (m^3/s).
- **Sediment rating curve:** $Q_s = a\,Q^b$ (b ≈ 1.5–3). Used for long-term sediment yield with the flow duration curve.
- Modern methods: turbidity meters, acoustic Doppler (ADCP), laser diffraction (LISST).

✎ **Exam Question:** Derive the Rouse equation for the distribution of suspended sediment concentration. What does the Rouse number indicate? (10 marks)

7. TOTAL SEDIMENT TRANSPORT

7.1 Methods

Total bed-material load ($q_t = q_b + q_s$) can be computed in two ways: (i) **separately** compute bed load and suspended load and add (Einstein, Bagnold, Van Rijn); (ii) **directly** compute the total load using a single formula related to the flow (Engelund–Hansen, Ackers–White, Yang).

7.2 Einstein's procedure (1950)

Combine the bed-load function (Section 5.4) with the suspended-load equation at reference level a = 2d; compute for each size fraction of the bed material, weight by its proportion i_b, and sum. Includes separation of the total hydraulic radius into R' (grain) and R'' (form). Requires iterations and charts. Accurate but laborious.

7.3 Engelund–Hansen (1967) – total load

Based on stream-power concept; used for sand-bed rivers (d > 0.15 mm, Fr < 1) with dunes:

$$g_t = 0.05\,\gamma_s\,V^2\,\sqrt{[\,d_{50}/((s-1)\,g)\,]} \times [\,\tau_0/((\gamma_s-\gamma)\,d_{50})\,]^{3/2}$$

g_t is the total load in weight per unit width per unit time (N/s/m); V is the mean velocity; τ_0 is the total bed shear stress (not corrected for form drag, which is built into the formula). Good for dune and plane-bed regimes.

7.4 Ackers–White (1973) and Yang (1973)

- **Ackers–White:** uses mobility number F_{gr} and dimensionless transport $G_{gr} = C(F_{gr}/A - 1)^m$, with coefficients that depend on D_* (for $D_* > 60$: C = 0.025, m = 1.5, A = 0.17). Calibrated for 0.04–4 mm flume data, Fr < 0.8.
- **Yang's unit stream power:** $\log C_t = 5.435 - 0.286 \log(wd/\nu) - 0.457 \log(u_*/w) + (1.799 - 0.409 \log(wd/\nu) - 0.314 \log(u_*/w)) \log(VS/w - V_{cr}S/w)$. C_t in ppm by weight. The sediment concentration is a function of the dimensionless unit stream power VS/w.
- **Brownlie, Van Rijn, Laursen** – field-validated alternatives.

7.4 Comparison of approaches

Formula	Type	Applies to
MPM	Bed load	Gravel and coarse sand, plane bed
Einstein	Bed + suspended	All sizes; needs bed-material gradation
Engelund–Hansen	Total load	Sand beds with dunes
Ackers–White	Total load	Sand and fine gravel
Yang	Total load (concentration)	Sand and gravel

7.5 Practical considerations

- Prediction accuracy is rarely better than a factor of 2; select a formula tested in conditions similar to your river and validate with measured data.
- Use mean annual sediment load = $\int Q_s(Q)\, f(Q)\, dQ$ with the flow duration curve. Most sediment moves during moderately high floods ("effective discharge", about bankfull flow).
- Sediment budget and continuity (**Exner equation**): $(1 - p)\,\partial z_b/\partial t + \partial q_t/\partial x = 0$. If the transport capacity decreases downstream ($\partial q_t/\partial x < 0$) deposition (**aggradation**) occurs; if it increases, scour (**degradation**).

8. DESIGN OF STABLE CHANNELS

8.1 Types of channels

- **Rigid-boundary (lined) channels** – no scour or deposition: designed by Manning's equation.
- **Erodible but non-silting channels (stable)** – the boundary is erodible but the channel does not change shape; designed by **permissible velocity** or **tractive force** methods.
- **Alluvial channels carrying sediment** – the channel adjusts its slope, width and depth: designed by **regime theory** (Kennedy, Lacey, Blench) or rational (Simons–Albertson) methods.

8.2 Maximum permissible velocity method (Fortier & Scobey)

Select a maximum velocity V_{max} that will not erode the channel bed and banks for the type of material.

Material	V_{max} (m/s), clear water
Fine sand	0.45
Sandy loam	0.55
Silt loam	0.60
Stiff clay	1.15
Fine gravel	0.75
Coarse gravel	1.20
Cobbles	1.50
Hard pan / soft rock	1.8 – 2.0

Procedure: choose n, side slopes m; compute R from $V = (1/n)R^{2/3}S^{1/2}$ with $V = V_{max}$, $A = Q/V$; select a bed width b and depth y that satisfy $A = (b + my)y$ and $R = A/P$.

8.3 Tractive force method (Lane, 1955)

The channel is stable if the shear stress on the bed and the banks is less than the critical shear (Section 3).

- Maximum shear stress on the **bed**: $\tau_b = \gamma\, y\, S_0$ (y = depth)
- Maximum shear stress on the **side slopes**: $\tau_s = 0.76\, \gamma\, y\, S_0$
- Permissible (critical) shear stress for side particle: $\tau_{c,s} = K\, \tau_{c,b}$, K as given in 3.4.

Design conditions:

$\gamma\, y\, S_0 \le \tau_{c,b}$ and $0.76\, \gamma\, y\, S_0 \le K\, \tau_{c,b}$ (so the controlling check is the side: $y \le K\, \tau_{c,b}/(0.76\, \gamma\, S_0)$)

For coarse non-cohesive material Lane's rule is $\tau_c \approx 0.75\, d_{75}$ (N/m^2, d_{75} in mm), consistent with Shields: fine sand ≈ 1 N/m^2; gravel with d = 25 mm ≈ 19 N/m^2.

8.4 Design steps for a trapezoidal stable channel

- 1. Choose side slope m (≥ angle of repose) and Manning n.
- 2. Choose bed shear limit τ_c and side slope reduction factor K.
- 3. Determine S_0 (given) and maximum allowable depth y from the side condition.
- 4. Choose b/y; compute A, R and Q from Manning. Adjust b until Q matches the design discharge.
- 5. Check V, Fr (< 0.8 to avoid instability), freeboard.

9. REGIME CHANNELS – KENNEDY AND LACEY THEORY

9.1 Regime concept

An alluvial channel carrying silt-laden water is in **regime** when, over a long period, the net scour and deposition are zero and the channel keeps constant width, depth and slope. Lacey defined three conditions for **true regime:** (i) channel flows uniformly in unlimited incoherent alluvium of the same character as that transported; (ii) discharge constant; (iii) silt grade and silt charge constant. In practice **initial regime** (bed only adjusted) or **final regime** (width also adjusted) is the best attained.

9.2 Kennedy's theory (1895)

From his observations on Upper Bari Doab canals in Punjab: **non-silting, non-scouring velocity** V_0 depends on depth y:

$$\mathbf{V_0 = 0.55\ m\ y^{0.64}}$$

m = **critical velocity ratio** (CVR) = V/V_0: m = 1 for ideal regime; m > 1 (> 1.1) scour; m < 1 deposition. Design: assume y, compute $V_0 = 0.55\ m\ y^{0.64}$, $A = Q/V_0$, then b from $A = (b + 0.5y)y$ (side slope 0.5H:1V), then S from Kutter's (or Manning's) formula; repeat until the CVR is about 1. It is a trial-and-error method; limitations: the relation takes no account of the silt charge or silt grade; width/depth relation is not obtained from the theory.

9.3 Lacey's theory (1930)

Based on data of stable canals in India and Egypt. Lacey considered the **silt factor f**:

$$\mathbf{f = 1.76\ \sqrt{d_{mm}}\ (d_{mm} = \text{mean grain size in millimetres})}$$

Lacey's regime equations (SI units):

$$\mathbf{V = (Q\ f^2 / 140)^{1/6}\ ;\ R = 5V^2 / (2f)}$$

$$\mathbf{P = 4.75\ \sqrt{Q}\ ;\ S = f^{5/3} / (3340\ Q^{1/6})}$$

and **A = Q/V**, which equals P·R for the regime section.

Other relations: wetted perimeter $P \propto \sqrt{Q}$ (width $\propto \sqrt{Q}$), $R \propto Q^{1/3}$, $S \propto Q^{-1/6}$, $V \propto Q^{1/6}\ f^{1/3}$.

Worked example: Design a regime channel for Q = 40 m^3/s, d = 0.3 mm (Lacey).

- f = 1.76 √0.3 = 0.964.
- $V = (40 \times 0.964^2/140)^{1/6} = (0.2655)^{1/6} = 0.80$ m/s.
- $A = Q/V = 50\ m^2$.
- $R = 5 \times 0.80^2/(2 \times 0.964) = 1.66$ m.
- P = 4.75 √40 = 30.0 m (check $A = P\ R = 49.8\ m^2$ ✓).
- $S = 0.964^{5/3}/(3340 \times 40^{1/6}) = 0.941/6176 = 1.52 \times 10^{-4} \approx 1$ in 6560.

9.4 Comparison of Kennedy and Lacey

Feature	Kennedy	Lacey
Basis	Silt-supporting eddies from the bed only	Eddies from both bed and sides; regime concept
Silt grade	Through CVR m only	Silt factor f
Section	Any width/depth, trial-and-error	Unique section for Q and f
Section shape	Trapezoidal	Semi-elliptical (regime) in theory
Equations	One (V_0 = 0.55 m $y^{0.64}$) + Kutter	Three main regime equations
Accuracy	Lower	Better (limited by data)

9.5 Blench and Simons–Albertson

- **Blench (1952):** introduces a bed factor F_b and a side factor F_s ($F_s \approx 0.1$ for slightly cohesive, 0.2 for medium, 0.3 for highly cohesive banks): width $b = \sqrt{(F_b Q/F_s)}$; depth $d = (F_s Q/F_b^2)^{1/3}$. Accounts for bank strength and sediment concentration.
- **Simons & Albertson (1963):** classify channel into 5 types by bed and bank material and sediment load; gives regime relations: $P = K_1 Q^{0.512}$, $R = K_2 Q^{0.361}$ etc. Based on data from Indian and American canals.
- **Modern rational (analytical) regime methods:** White–Bettess–Paris, Parker's gravel-bed river theory, extremal hypotheses (minimum stream power, maximum transport).

✎ **Exam Question:** Design an irrigation channel by Lacey's theory for Q = 30 m^3/s, silt factor 1.0. *Ans: $V = (30\times1/140)^{1/6} = 0.774$ m/s; $A = 38.8\ m^2$; $R = 5\times0.599/2 = 1.50$ m; $P = 4.75\sqrt{30} = 26.0$ m; $S = 1/(3340\times30^{1/6}) = 1/5885 = 1.7\times10^{-4}$.*

10. RESERVOIR SEDIMENTATION AND TRAP EFFICIENCY

10.1 The problem

A dam interrupts the river flow. The reduced velocity in the reservoir causes the coarse fractions to settle at the head (**delta**) and fine fractions to settle further in (**bottom sets**) or be carried as **turbidity (density) currents** along the old river bed to the dam. Deposition occupies storage capacity: dead, live and flood-control storage, reduces energy and water supply, raises upstream flood levels (backwater), damages turbines and gates (abrasion) and starves the downstream river of sediment (degradation).

Worldwide, reservoirs lose about 0.5–1 % of storage per year; in India the rate is 0.5–1.5 % and in the Himalayan region more.

10.2 Sediment yield from the catchment

- **Measured:** sediment rating curve + flow duration curve; reservoir surveys (hydrographic, capacity surveys with echo-sounders).
- **Empirical: USLE** A = R K L S C P (soil loss in t/ha/yr; R rainfall erosivity, K soil erodibility, LS slope length and steepness, C cover, P practice); **MUSLE** for single storms; sediment delivery ratio (SDR) accounts for the proportion reaching the outlet; **Khosla's and Joglekar's** empirical formulae for Indian catchments relate annual sediment yield to catchment area.
- Typical yields: 50–500 t/km^2/yr (temperate) to > 2000 t/km^2/yr (Himalayan, loess).

10.3 Trap efficiency

Trap efficiency (TE) is the proportion of the incoming sediment that is deposited in the reservoir:

$$\text{TE} = (\text{sediment inflow} - \text{sediment outflow})/\text{sediment inflow} \times 100\ \%$$

Depends on: capacity/inflow ratio, size and settling velocity of particles, reservoir shape, operation (flushing, sluicing), detention time.

Brune's curve (1953): empirical relation between TE and the capacity–inflow ratio C/I (C = reservoir capacity, I = mean annual inflow volume), derived from 40 normal-ponded reservoirs. Approximate values from the median curve:

C/I	TE (approx.)
1.0	≈ 97 %
0.1	≈ 90 %
0.01	≈ 55–70 %
0.001	≈ 10–30 %

Brune's curves: upper envelope – coarse sediment; median – normal; lower envelope – fine, colloidal sediments.

Churchill (1948): TE is a function of the **sedimentation index** SI = (retention period)/(mean velocity through the reservoir); TE falls as SI decreases, i.e. as water passes through faster. Mainly for small, short-detention reservoirs.

Camp (1945) / Camp–Dobbins (settling basin) concept: an ideal settling basin removes all particles with $w \geq Q/A_s$ (overflow rate = fall velocity). The fraction removed of each class = $w/(Q/A_s)$ for $w < Q/A_s$. TE of whole reservoir = Σ(size fraction × removal ratio).

10.4 Estimating the reservoir life

Annual sediment volume deposited $V_s = (TE \times W_s)/\rho_{deposit}$, with W_s = annual sediment load (t/yr) and $\rho_{deposit}$ = dry density of deposit (t/m^3; sand 1.5; silt 1.2; clay 0.7 – 1.0; or from **Lara–Pemberton** formula for compaction with time).

Example: capacity 100 Mm^3; sediment inflow 1.5 Mt/yr; TE = 90 %; deposit density = 1.2 t/m^3. V_s = 0.9 × 1.5/1.2 = 1.125 Mm^3/yr, so the **half-life (50 % filled)** ≈ 44 years and complete filling ≈ 89 years (without allowing for decreasing TE as capacity drops).

10.6 Control and management of reservoir sedimentation

Category	Measures
Reduce sediment inflow	Catchment treatment (afforestation, contour bunds, terracing, check dams), sediment-trap dams, vegetative filter strips
Route sediment past the reservoir	Bypass channel or tunnel, off-stream reservoirs, **sluicing** (open gates during floods)
Pass sediment through	**Venting of turbidity currents** through low-level outlets
Remove deposited sediment	**Hydraulic flushing** (draw-down, needs a large low-level outlet), dredging (hydraulic/mechanical), dry excavation
Adaptive strategies	Raise the dam; increase storage upstream; design with extra dead storage (e.g. 50-year sediment allowance); sediment-resistant intakes; trap-efficiency monitoring

★ **Key Point:** Always include a sediment-storage allowance (usually for 50–100 years) in a reservoir's design, and place the power intake and outlets above the expected sediment level.

10.7 Quick revision – formulae

Topic	Formula
Stokes' fall velocity	$w = (s-1)g\,d^2/18\nu$
Shields parameter	$\tau_* = \tau_0/[(\gamma_s - \gamma)d]$; $\tau_{*c} \approx 0.045$
Shear velocity	$u_* = \sqrt{(gRS)}$
MPM	$\Phi = 8(\tau_*' - 0.047)^{3/2}$
Rouse equation	$C/C_a = [(h-y)a/(y(h-a))]^Z$, $z = w/\kappa u_*$
Engelund–Hansen	$g_t = 0.05\gamma_s V^2\sqrt{(d/(s-1)g)}\,\tau_*^{3/2}$
Lacey	$V = (Qf^2/140)^{1/6}$; $R = 5V^2/2f$; $P = 4.75\sqrt{Q}$; $f = 1.76\sqrt{d_{mm}}$
Kennedy	$V_0 = 0.55\ m\ y^{0.64}$
Trap efficiency	Brune curve: TE = f(C/I)

✎ **Exam Question:** What is trap efficiency of a reservoir? Explain the factors influencing it and describe methods to control reservoir sedimentation. (10 marks)

23A01604b • OPEN CHANNEL FLOW

UNIT – 5

Flow Measurements and Hydraulic Modeling

Complete Study Notes • Concepts · Derivations · Formulae · Exam Questions

Syllabus Topics

- Sharp-Crested Weirs, Broad-Crested Weirs, Critical Depth Flumes
- Recent Advancements in Open Channel Flow Measurements
- Physical Modelling in Hydraulics
- Dimensional Analysis
- Modelling Closed Flows and Free Surface Flows
- Distorted Models
- Design of Physical Models

1. FLOW MEASUREMENT IN OPEN CHANNELS – OVERVIEW

1.1 Why measure flow?

Discharge data are needed for water allocation and billing in irrigation canals, flood forecasting, design of hydraulic structures, reservoir operation, environmental flow assessment and calibration of hydraulic models.

1.2 Methods of measuring discharge

Method	Principle	Typical use
Velocity–area	$Q = \Sigma v_i a_i$ from current-meter readings	Rivers and large canals
Hydraulic structures	Unique head–discharge relation (weirs, flumes)	Canals, labs, small streams
Dilution (tracer) gauging	Dilution of injected tracer: $Q = q(C_1 - C_2)/(C_2 - C_0)$	Turbulent mountain streams
Slope–area	Manning's equation using surveyed slope and section	Post-flood peak estimates
Acoustic / electromagnetic	Doppler shift, transit-time, induced voltage	Continuous, automated stations
Float / surface-velocity methods	Float time × surface-to-mean factor (0.85)	Rough estimates

1.3 Velocity–area method in brief

The cross-section is divided into verticals (spacing about 1/20 to 1/15 of width). At each vertical the mean velocity is taken as velocity at 0.6 d (one-point method), or the average at 0.2 d and 0.8 d (two-point method), or the average of 0.2, 0.6 and 0.8 d. Partial discharge $q_i = v_i \times d_i \times w_i$; total $Q = \Sigma q_i$ (mid-section method). A **rating curve** $Q = C\,(h - h_0)^b$ is then developed at the gauging station, where h is the stage and h_0 the gauge height at zero flow.

1.4 Hydraulic-structure approach

The idea is to force the flow through a **control section** where the flow passes critical depth (or a defined head/discharge relation). Then Q depends on only one measured quantity – the upstream head H. Examples: weirs (thin-plate, broad-crested), flumes (Venturi, Parshall, long-throated). Their advantage is high accuracy (± 2–5 %) and simplicity; disadvantage is head loss and possible sediment deposition.

★ **Key Point:** A measuring structure must satisfy three conditions: (i) a well-defined control (critical flow or free nappe), (ii) a calm, straight approach channel with uniform velocity distribution, and (iii) head measured at the correct location (upstream, about 3–4 H_{max} from the weir, in a stilling well if needed).

2. SHARP-CRESTED WEIRS

2.1 Description

A **sharp-crested (thin-plate) weir** is a vertical plate with a sharp upstream edge (bevelled 45° downstream, crest thickness ≤ 2 mm) placed across the channel normal to the flow. Water passes over the crest in a **nappe** that springs free from the plate. The weir may be rectangular, triangular (V-notch), trapezoidal (Cipolletti) or proportional (Sutro).

Requirements: crest level above the bed (weir height P) ≥ $2H_{max}$ and ≥ 0.3 m; head H ≥ 0.06 m (surface-tension effects are important at smaller head); H ≤ 0.6 m; channel width ≥ 2 H; the nappe

must be **fully aerated** (vented), downstream water level below crest (free flow).

2.2 Rectangular weir – derivation

Consider a horizontal strip of thickness dh at depth h below the free surface, width L. Velocity at the strip $\sqrt{(2gh)}$. Discharge through the strip: $dQ = C_d\, L\, \sqrt{(2gh)}\, dh$. Integrating from 0 to H:

$$\mathbf{Q = (2/3)\, C_d\, L\, \sqrt{(2g)}\, H^{3/2}}$$

This is the basic formula. Typical $C_d = 0.60–0.62$. Rehbock's formula: $C_d = 0.602 + 0.083\, H/P$ (equivalently $Q = 1.78\, L\, H^{3/2}(1 + 0.14\, H/P)$ in SI).

End contractions: if the weir does not span the full channel width, the nappe contracts at the sides and the effective length is reduced (**Francis formula**):

$$\mathbf{Q = 1.84\, (L - 0.1\, n\, H)\, H^{3/2}}$$ **(SI; n = number of end contractions: 2 for fully contracted, 0 for suppressed weir)**

Velocity of approach: when $V_a = Q/(B(P + H))$ is not negligible, use the total head $H + h_a$, $h_a = V_a^2/2g$:

$$\mathbf{Q = 1.84\, L\, [\, (H + h_a)^{3/2} - h_a^{3/2}\,]}$$ **(suppressed weir)**

2.3 Triangular (V-notch) weir

Used for **small discharges** (< 30 L/s) because a small change in Q produces a large change in H, so it is more sensitive than the rectangular weir. For a notch of angle θ:

$$\mathbf{Q = (8/15)\, C_d\, \sqrt{(2g)}\, \tan(\theta/2)\, H^{5/2}}$$

For a 90° notch ($C_d = 0.58$): $\mathbf{Q \approx 1.38\, H^{5/2}}$ (SI). For 60° notch: $Q \approx 0.80\, H^{5/2}$. Range of H: 0.06–0.38 m.

2.4 Trapezoidal (Cipolletti) weir

Side slopes 1 horizontal : 4 vertical. The extra area at the sides compensates for the end contraction loss, so the discharge is the same as the suppressed rectangular weir of the same crest length:

$$\mathbf{Q = 1.86\, L\, H^{3/2}}$$ **(SI)**

2.6 Submerged (drowned) weir

When the tailwater rises above the crest, the discharge reduces. **Villemonte (1947):**

$$\mathbf{Q_s = Q_1\, [\, 1 - (H_2/H_1)^n\,]^{0.385}}$$

where Q_1 is the free-flow discharge for the upstream head H_1, H_2 = downstream head over the crest, and n = 1.5 (rectangular), 2.5 (V-notch).

2.7 Worked example

A rectangular, fully contracted sharp-crested weir has L = 1.2 m and a measured head H = 0.25 m. Find the discharge using the Francis formula.

- $L' = L - 0.1\, n\, H = 1.2 - 0.1 \times 2 \times 0.25 = 1.15$ m.
- $H^{3/2} = 0.25^{1.5} = 0.125$.

- Q = 1.84 × 1.15 × 0.125 = **0.264 m^3/s**.

For a 90° V-notch with H = 0.20 m: Q = 1.38 × $0.20^{2.5}$ = 1.38 × 0.0179 = 0.0247 m^3/s ≈ 25 L/s.

2.8 Errors in measurement

Error in Q due to error in H: for rectangular weir dQ/Q = 1.5 dH/H; for V-notch dQ/Q = 2.5 dH/H. A 1 % error in H gives a 1.5 % (rectangular) or 2.5 % (triangular) error in Q. Therefore the head must be measured carefully (point gauge or float in stilling well, accuracy ± 0.3 mm).

✎ **Exam Question:** Derive the discharge formula for a V-notch weir. Why is it preferred to a rectangular weir for small discharges? (8 marks)

3. BROAD-CRESTED WEIRS

3.1 Description

A **broad-crested weir** is a raised, flat-topped obstruction across the channel whose crest length L in the flow direction is large enough for the streamlines on the crest to be parallel and the pressure hydrostatic. Flow passes through critical depth on the crest, which is the control section. The upstream face is rounded or bevelled to avoid separation.

Limits of application: 0.08 < H_1/L < 0.5 (H_1 = upstream head above the crest). If H_1/L < 0.08, boundary friction dominates; if H_1/L > 0.5 the weir behaves as a **short-crested weir** (curvature of streamlines is significant). Typical L = 0.5–3 m for field weirs.

3.2 Theory

Consider a rectangular weir and a free flow (no tailwater influence). Let H be the upstream head above the crest (measured at ≥ 3H upstream). Neglect the velocity of approach and losses. Applying Bernoulli between the upstream section and the crest:

H = $y_c + V_c^2/2g$. For critical flow $V_c^2/2g = y_c/2$, so

y_c = (2/3) H and $V_c = \sqrt{(g\, y_c)} = \sqrt{(2gH/3)}$

$q = y_c V_c$ = (2/3) H $\sqrt{(2gH/3)}$:

$$\mathbf{Q = (2/3)^{3/2}\ \sqrt{g}\ L\ H^{3/2} = 1.705\ L\ H^{3/2}}$$

Including a **discharge coefficient** C_d (for friction and curvature) and the velocity-of-approach coefficient C_v:

$$\mathbf{Q = 1.705\ C_d\ C_v\ L\ H^{3/2}\ (SI)}$$

with C_d ≈ 0.85–1.0 for well-built weirs (Bos recommends C_d = 0.93 + 0.10 H/L for 0.1 < H/L < 0.7, approximately). The discharge is proportional to $H^{3/2}$ and **independent of tailwater** provided the weir is **modular**.

3.3 Modular limit

Free (modular) flow exists when the critical depth forms on the crest; the tailwater level (head H_2 above crest) must not exceed a limit:

H_2/H_1 ≤ 0.66–0.75 (modular limit). For larger submergence the flow is **drowned** and Q is reduced; correction for submergence is required (Tailwater correction factor, or two-head measurement).

3.5 Comparison: sharp-crested vs broad-crested

Feature	Sharp-crested	Broad-crested
Accuracy	High (± 1–3 %)	Moderate (± 3–5 %)
Head loss	Large	Small to moderate
Sediment / debris	Collects sediment (poor)	Passes sediment better
Strength	Fragile (plate)	Robust (concrete)
Q–H relation	$Q \propto H^{3/2}$ (rect.), $H^{5/2}$ (V)	$Q \propto H^{3/2}$
Control section	Free nappe	Critical depth on crest
Typical use	Laboratory, small canals	Field channels, irrigation

3.6 Worked example

A broad-crested weir (L_{crest} = 2 m in the flow direction, width b = 5 m) is built in a canal. Head measured above the crest H = 0.60 m. Take C_d = 0.92 and C_v = 1.0.

- Check H/L = 0.30 (within 0.08–0.5 ✓).
- $Q = 1.705 \times 0.92 \times 5 \times 0.60^{1.5} = 1.705 \times 0.92 \times 5 \times 0.4648 =$ **3.65 m^3/s**.
- y_c = 0.40 m; V_c = 1.98 m/s; the downstream water depth above the crest must be < 0.66 × 0.60 = 0.40 m for modular flow.

✎ **Exam Question:** Derive the discharge equation for a broad-crested weir and explain the concept of modular limit. (10 marks)

4. CRITICAL-DEPTH FLUMES

4.1 Principle

A **flume** is a specially shaped channel section in which the width is contracted and/or the bed raised so that the flow passes through critical depth in the throat. Compared to weirs, flumes cause **less head loss**, pass sediment and debris, and need no stilling pool. Because critical flow prevails in the throat, the discharge is a unique function of the upstream head (provided submergence is below the modular limit).

4.2 Venturi flume (standing-wave flume)

Contraction in width (and sometimes bed hump). For a rectangular throat of width b_t with critical depth $y_c = (2/3)H$:

$$\mathbf{Q = C_d\,(2/3)^{3/2}\,\sqrt{g}\,b_t\,H^{3/2} = 1.705\,C_d\,b_t\,H^{3/2}}$$

Here H = upstream total energy head above the throat bed. $C_d \approx 0.95$–0.99 (affected by boundary-layer growth). A gradual converging section (typically 1:3 to 1:5) minimises energy loss; a diverging section (1:6) recovers some head.

4.3 Parshall flume

Developed by R. L. Parshall (1920s, Colorado). Consists of a **converging section**, a **throat** with a downward sloping floor and a **diverging section** with an upward sloping floor. Size is designated by the throat width W (from 1 in to 50 ft). Two stilling wells measure heads: H_a (in converging section at 2/3 of its length) and H_b (at the throat).

Free-flow discharge (US units: W and H_a in feet, Q in ft^3/s):

$$Q = 4\ W\ H_a^{1.522\ W^{0.026}}$$

Example: W = 1 ft, H_a = 0.5 ft: Q = 4 × $0.5^{1.522}$ = 1.39 ft^3/s ≈ 39 L/s. (SI rating tables are available for every standard size.) Submergence S = H_b/H_a; free flow if S is below the transition limit: **50 % (W = 1–3 in), 60 % (6–9 in), 70 % (1–8 ft), 80 % (10–50 ft)**. For higher submergence a correction (from charts) is applied and both H_a and H_b are needed.

Advantages: works with free and submerged flow; small head loss (about 1/4 of a weir); self-cleaning; insensitive to approach velocity; wide range of sizes (1 to 1000s L/s).
Disadvantages: complicated geometry, needs precise construction; high cost; relatively sensitive to changes from standard dimensions.

4.5 Long-throated flumes (Bos, Clemmens, Replogle)

A flume with a throat long enough (L/H > 1–3) that the streamlines are parallel and hydrostatic and critical depth forms in the throat. The **rating can be computed theoretically** for any shape (rectangular, trapezoidal, circular, U-shape) with the energy equation and boundary-layer correction, so no laboratory calibration is needed:

$$Q = C_d\ A_c\ \sqrt{(g\ A_c/T_c)},\ H_1 = y_c + A_c/(2T_c)\ (A_c,\ T_c \text{ at critical depth})$$

The program **WinFlume** (USBR/ILRI) designs the flume and the rating table. These flumes are standard for irrigation turnouts.

4.6 Choice of measuring structure

Condition	Preferred device
Very small flows (< 30 L/s), accurate lab work	90° or 60° V-notch weir
Medium flows, clean water	Rectangular sharp-crested weir
Sediment-laden canals, small head available	Flume (Parshall, cutthroat, long-throated)
Field irrigation canals, robust, low maintenance	Broad-crested or Crump weir; RBC flume
Wide variation in flow and unknown submergence	Parshall flume

★ **Key Point:** Modular limit rule: weirs and flumes work best when the downstream water level is low enough that flow is "free". Always check submergence under the maximum flow.

5. RECENT ADVANCEMENTS IN OPEN-CHANNEL FLOW MEASUREMENT

5.1 Limitations of conventional methods

Structures need head loss and maintenance; current-meter gauging is slow, labour-intensive, risky in floods and cannot give continuous records. Modern technologies provide non-intrusive, rapid, automated and remote measurements.

5.2 Acoustic Doppler technologies

- **Acoustic Doppler Current Profiler (ADCP):** transmits sound pulses along 3–4 beams and measures the Doppler shift of echoes from suspended particles to give velocity profiles in cells through the depth. Moving-boat ADCP (e.g. on a tethered or remote-controlled boat) measures a whole cross-section in 5–15 minutes, including bottom-tracking to measure boat velocity and depth; Q = ∫∫ v·n dA. Used widely by USGS, CWC etc. Fixed (horizontal) H-ADCP mounted on the bank gives **index-velocity** rating for continuous discharge.

- **Acoustic Doppler Velocimeter (ADV):** high-frequency point measurement of 3-D velocity (to 200 Hz) in laboratory and field; gives turbulence characteristics and Reynolds stresses.
- **Ultrasonic transit-time (acoustic path) flow meters:** sound pulses are sent in both directions across the channel along one or more paths; the time difference gives the path-average velocity; used in power canals, large rivers and tidal channels.
- **Ultrasonic level sensor + weir/flume:** non-contact level measurement for automatic head recording.

5.3 Radar and optical (non-contact) methods

- **Surface-velocity radar:** Doppler radar (K- or X-band) mounted on a bridge or a bank measures the surface velocity; the mean velocity $V = \alpha V_s$ ($\alpha \approx 0.85$) is used with surveyed cross-section to give Q. Safe in floods.
- **Radar level sensors** for stage.
- **Image-based methods: LSPIV** (large-scale particle image velocimetry), **PTV** and **STIV** (space-time image velocimetry) track natural surface features (foam, ripples) or seeding in video from fixed cameras or **UAV/drones**, giving 2-D surface velocity fields. Low cost and no contact; calibrated by ground control points.
- **Satellite and UAV remote sensing:** width and water-surface slope from satellite altimetry (e.g. SWOT), Landsat/Sentinel imagery for river width; Q estimated from at-many-stations hydraulic geometry (AMHG).

5.5 Digital and smart-hydrology developments

- **IoT telemetry:** loggers send stage, velocity and discharge in real time via GSM/satellite/LoRa to SCADA systems; used for canal automation and flood warning.
- **Machine learning and data assimilation:** ANN, random forests and LSTM networks to derive rating curves, fill gaps, and estimate discharge from camera images.
- **Smart/automatic gates (Constant-downstream-level, Baffle modules)** that measure and control flow simultaneously (e.g. flume gates and constant-downstream-level gates).
- **Standards:** ISO 748 (velocity–area), ISO 1438 (thin-plate weirs), ISO 4359 (flumes), ISO 6416 (ultrasonic velocity method), ISO 24578 (acoustic Doppler profilers).

5.6 Comparison of modern techniques

Technique	Advantages	Limitations
Moving-boat ADCP	Fast, whole cross-section, safe	Bed-load error, expensive, needs minimum depth
H-ADCP / index-velocity	Continuous Q, backwater ok	Needs calibration
Surface-velocity radar	Non-contact, safe in floods	Surface only; wind effects
LSPIV / drone	Low cost, 2-D flow field	Needs seeding/features, lighting
Ultrasonic transit time	Accurate (± 1–2 %)	Needs stable channel, installation cost
Tracer dilution	No cross-section needed	Needs good mixing length

✎ **Exam Question:** Discuss recent advancements in open-channel flow measurement and compare ADCP and radar methods. (10 marks)

6. DIMENSIONAL ANALYSIS

6.1 Purpose

Dimensional analysis expresses a physical relation in terms of **dimensionless groups**. It reduces the number of variables, guides experiments, and forms the basis for model–prototype similarity. Fundamental dimensions: mass **M**, length **L**, time **T** (and temperature if needed).

6.3 Rayleigh's method

Write the dependent variable as a product of powers of independent variables, then equate indices of M, L, T. It is suitable for ≤ 4 variables. Example: discharge over a rectangular weir $Q = K\,L^a\,H^b\,g^c$ → dimensions $L^3T^{-1} = L^{a+b+c}\,T^{-2c}$ → c = 1/2 and a + b = 5/2. Since discharge is proportional to the width, a = 1 and b = 3/2, giving $\mathbf{Q = K\,L\,H^{3/2}\,g^{1/2}}$, the same form as the weir formula.

6.4 Buckingham π theorem

If a physical phenomenon involves n variables and m fundamental dimensions, it can be expressed as a relationship between **(n − m) independent dimensionless π terms**. Procedure:

- 1. List all variables (n) affecting the phenomenon, with their dimensions.
- 2. Count fundamental dimensions (m); number of π terms = n − m.
- 3. Select m **repeating variables** that (a) together contain all fundamental dimensions, (b) cannot form a dimensionless group among themselves – commonly one geometric (D or L), one kinematic (V) and one dynamic (ρ).
- 4. Form each π by multiplying one non-repeating variable by the repeating variables with unknown exponents, and solve exponents so that the product is dimensionless.
- 5. Express the result as $\pi_1 = \varphi(\pi_2, \pi_3, \ldots)$. Combine groups if needed (product/ratio) to get standard numbers.

Example – pressure drop in a pipe. Δp = f(ρ, V, D, μ, L, ε): n = 7, m = 3, so 4 π-terms. Choose repeating variables ρ, V, D:

- $\pi_1 = \Delta p/(\rho V^2)$ (Euler number)
- $\pi_2 = \mu/(\rho VD) = 1/Re$
- $\pi_3 = L/D$, $\pi_4 = \varepsilon/D$

So $\Delta p/(\rho V^2) = \varphi(Re, L/D, \varepsilon/D)$ → $h_f = f\,(L/D)(V^2/2g)$ – the Darcy–Weisbach equation, with $f = \varphi(Re, \varepsilon/D)$.

6.5 Important dimensionless numbers

Number	Expression	Ratio of forces	Governs
Reynolds	$Re = \rho VL/\mu$	Inertia : viscous	Pipe flow, boundary layer, flow around bodies
Froude	$Fr = V/\sqrt{(gL)}$	Inertia : gravity	Open-channel flow, spillways, waves, ship resistance
Euler	$Eu = \Delta p/(\rho V^2)$	Pressure : inertia	Pressure drops, cavitation ($\sigma = (p - p_v)/(0.5\rho V^2)$)
Weber	$We = \rho V^2 L/\sigma$	Inertia : surface tension	Small-scale free-surface flow, droplets, thin nappes
Mach / Cauchy	$Ma = V/c$	Inertia : elastic	Compressible flow, water hammer
Strouhal	$St = fL/V$	Unsteady : inertia	Vortex shedding

7. PHYSICAL MODELLING IN HYDRAULICS

7.1 Why physical models?

Many hydraulic problems are too complex for theory alone: 3-D flow around piers and intakes, energy dissipators, spillways, harbours, river training, sediment transport, cavitation and air entrainment. A **physical (scale) model** is a reduced (or sometimes enlarged) replica built to reproduce the flow, in which measurements are made and then converted to prototype values. Numerical (CFD) models are increasingly used, but a physical model remains the most reliable for complex free-surface flow and for validating numerical codes.

Terminology: **prototype** = the actual structure; **model** = replica; **scale ratio** (X_r) = prototype value / model value (e.g. $L_r = L_p/L_m$). In these notes we use model/prototype ratio $X_r = X_m/X_p$; both conventions occur, so check the question.

7.2 Similitude (similarity) conditions

- **Geometric similarity:** all linear dimensions in a constant ratio L_r; areas $A_r = L_r^2$; volumes L_r^3. Includes roughness, i.e. $\varepsilon_r = L_r$.
- **Kinematic similarity:** velocities and accelerations at corresponding points are in a constant ratio (streamline patterns are similar).
- **Dynamic similarity:** ratio of all forces (inertia, gravity, viscous, pressure, surface tension, elastic) is the same in model and prototype. If dynamic similarity is satisfied with geometric similarity, kinematic similarity follows.

Dynamic similarity requires equality of the relevant dimensionless numbers. Since it is impossible to equate Re, Fr and We simultaneously with the same fluid, the **dominant force** is identified and the corresponding number is equated; other forces are controlled by choosing model size so that their effects are negligible (scale effects).

8. MODELLING OF CLOSED-CONDUIT FLOWS AND FREE-SURFACE FLOWS

8.1 Closed-conduit (pressurised) flows – Reynolds model law

In pressure flows (pipes, valves, turbines, pumps, bends), viscous and inertia forces dominate while gravity only affects the hydrostatic pressure, which does not influence the flow pattern. Hence equality of the **Reynolds number**:

$$(\rho V L/\mu)_m = (\rho V L/\mu)_p \rightarrow V_r\, L_r/\nu_r = 1$$

Scale ratios (Reynolds model, subscript r = model/prototype):

Quantity	Ratio	Same fluid ($\nu_r = 1$)
Length	L_r	L_r
Velocity	$V_r = \nu_r/L_r$	$1/L_r$
Time	$T_r = L_r^2/\nu_r$	L_r^2
Discharge	$Q_r = \nu_r\, L_r$	L_r
Force	$F_r = \rho_r\, \nu_r^2$	1
Pressure	$p_r = \rho_r\, \nu_r^2/L_r^2$	$1/L_r^2$

Practical problem: with the same fluid (water) in a small model, the velocity would need to be larger by $1/L_r$, which is often impractically high (e.g. a 1:10 model needs 10 × prototype velocity).

Remedies: use a different fluid (e.g. air, oil with lower ν), pressurise or heat the water (reduces ν), or rely on **Reynolds-number independence** – above a critical Re (fully rough turbulent flow) friction factor is independent of Re, so only geometric (including roughness) and a "sufficiently large" Re are required. For pipe flow it is enough that the model Re > 10^4.

Cavitation models also equate the cavitation number $\sigma = (p - p_v)/(0.5\rho V^2)$. **Water-hammer models** use the Cauchy (Mach) number – scale a, V and time with L/a.

8.2 Free-surface flows – Froude model law

In open channels, spillways, flow over weirs, river and harbour flows, gravity is the governing force, and viscous effects are secondary. Therefore **Froude number** must be equal:

$$\mathbf{(V/\sqrt{(gL)})_m = (V/\sqrt{(gL)})_p \rightarrow V_r = \sqrt{(L_r)}\ (g_r = 1)}$$

Scale ratios (Froude model):

Quantity	Ratio	Exponent of L_r
Length	L_r	1
Area	A_r	2
Volume		3
Velocity	V_r	1/2
Time	$T_r = L_r/V_r$	1/2
Acceleration		0
Discharge	$Q_r = A_r V_r$	5/2
Force	$F_r = \rho_r L_r^3$	3
Pressure	$p_r = \rho_r L_r$	1
Power	$P_r = F_r V_r$	7/2
Manning's n	$n_r = L_r^{1/6}$	1/6
Kinematic viscosity (if scaled)	$\nu_r = L_r^{3/2}$	3/2

Worked example: A spillway has a design discharge of 4500 m^3/s. A Froude model of scale 1:50 (L_r = 1/50) is built.

- $Q_m = 4500 \times (1/50)^{5/2} = 4500/17\,678 =$ **0.255 m^3/s**.
- A prototype velocity of 20 m/s corresponds to $V_m = 20/\sqrt{50} = 2.83$ m/s.
- A prototype flood of 12 h lasts $12/\sqrt{50} = 1.70$ h in the model.
- A measured model force of 20 N corresponds to a prototype force = $20 \times 50^3 = 2.5$ MN ($\rho_r = 1$).
- A measured head of 0.12 m in the model = 6.0 m in the prototype.

8.3 Scale effects

Because water is used in both model and prototype, **viscosity and surface tension are not scaled** correctly in a Froude model. The model Reynolds number is lower than prototype by $L_r^{3/2}$, so friction, boundary-layer thickness, air entrainment and wave breaking are not reproduced exactly. To limit scale effects:

- Choose a model large enough so that **flow depth ≥ 3 cm**, **head over the crest ≥ 5 cm** (We effect), **Re ≥ 2000–5000** (turbulent flow in model), and model channel R ≥ ~1–2 cm.
- Adjust roughness so that the Froude-scaled $n_r = L_r^{1/6}$ is obtained.
- For **air entrainment**, model scales larger than 1:15 are preferred; extrapolate results if needed.
- **Cavitation** cannot be modelled by a Froude model – use a separate, low-pressure test.

★ **Key Point:** Typical Froude-model scales: spillways and energy dissipators 1:20 to 1:60; intake/vortex studies 1:5 to 1:25; harbours 1:100 to 1:150 horizontally; rivers 1:100 to 1:500 horizontally (distorted).

✎ **Exam Question:** A 1:25 Froude model of a spillway has a discharge of 0.5 m^3/s. Find the prototype discharge and the ratio of the forces. *Ans:* $Q_p = 0.5 \times 25^{5/2} = 0.5 \times 3125 = 1562\ m^3/s$; $F_p/F_m = 25^3 = 15\ 625$.

9. DISTORTED MODELS

9.1 What is a distorted model?

A **distorted model** is one in which the **horizontal scale X_r and vertical scale Y_r are different**. The **distortion ratio** (distortion factor) is

$n_d = Y_r/X_r$ (vertical scale ratio ÷ horizontal scale ratio)

with scale ratios taken as model/prototype. Example: horizontal 1:400 and vertical 1:40 give n_d = (1/40)/(1/400) = 10.

9.2 Why distort?

- Large rivers/estuaries/harbours have lengths of kilometres and depths of metres. An undistorted model at 1:1000 would have depths of millimetres: surface tension and laminar flow, very high viscous effects, and measuring accuracy is lost.
- Need to reproduce **sufficient bed shear to move sediment**, and keep **turbulent flow (Re > 2000)**.
- Fit the model within the available laboratory space.
- Get enough **slope** for gravity-driven flow in the model.

9.3 Scale relationships for distorted Froude models

Let X_r = horizontal scale ratio and Y_r = vertical scale ratio (model/prototype). In a distorted model $Y_r > X_r$, i.e. the vertical is exaggerated.

Quantity	Relation	Derivation
Velocity	$V_r = \sqrt{(Y_r)}$	Froude number (depth as length scale)
Discharge	$Q_r = X_r Y_r^{3/2}$	Q = width × depth × velocity
Time (horizontal movement)	$T_r = X_r/\sqrt{Y_r}$	t = distance/velocity
Slope	$S_r = Y_r/X_r$	vertical ÷ horizontal
Hydraulic radius	$R_r \approx Y_r$ (wide channel)	R ≈ y
Manning's n	$n_r = Y_r^{2/3}/X_r^{1/2}$	$V = R^{2/3}S^{1/2}/n$
Time (vertical movement, e.g. settling)	$Y_r/V_{r,v}$	Depends on fall velocity

Example: A river model has X_r = 1/400 and Y_r = 1/40. Distortion = 10. $Q_r = (1/400)(1/40)^{3/2}$ = 1/(400 × 253) = 1/101 192. So a prototype flood of 5000 m^3/s corresponds to 0.049 m^3/s. V_r = 1/6.32 (a prototype 3 m/s becomes 0.47 m/s). T_r = (1/400)/(1/6.32) = 1/63.2: a 24 h prototype event lasts 22.8 min. Slope in model = 10 × prototype slope. $n_r = (1/40)^{2/3}/(1/400)^{1/2}$ = 0.0852 × 20 = 1.70, so the model roughness must be 1.7 times the prototype n – use artificial roughness (rods, wire mesh, gravel) if the smooth model bed is not rough enough; if less roughness is needed,

polish the bed.

9.4 Limitations of distortion

- Velocity distribution and secondary currents (bends) are different; **3-D effects** and eddies are not scaled correctly.
- Side-slope and bank angles are changed: steeper banks → different stability and scour.
- **Wave patterns** and refraction in harbours are affected; vertical accelerations become important; hydrostatic assumption can fail.
- Sediment transport: grain size, fall velocity and shear stress must be re-scaled for the movable bed; time scales for bed-load and suspended load are different.
- Limit distortion to about 5–10 (maximum ≈ 20) for rivers; use undistorted models for local structures (bridge piers, intake).

★ **Key Point:** Rule of thumb: distort only the minimum necessary; avoid distortion where flow patterns (separation, eddies, local scour) are important.

10. DESIGN OF PHYSICAL MODELS

10.1 Step-by-step procedure

- **Step 1 – Define objectives.** What will the model show: discharge capacity, pressures, water-surface profile, scour, energy dissipation, forces, sediment movement? Choose the dominant forces (gravity → Froude; viscosity → Reynolds; surface tension → Weber).
- **Step 2 – Select the type of model.** Rigid or movable bed; undistorted or distorted; sectional or comprehensive; fixed or tidal.
- **Step 3 – Select the scale.** Compromise between (a) space and discharge available in the laboratory, (b) cost and construction time, (c) measurement accuracy, (d) scale-effect limits (minimum depth ≈ 3 cm; weir head ≥ 5 cm; Re ≥ 2000; We effects small).
- **Step 4 – Derive scale ratios** for all variables (Section 8, 9), and calculate model discharge, velocities, depths, slopes, roughness, time scale.
- **Step 5 – Layout and construction.** Provide an approach channel (≥ 5–10 channel widths, with flow straighteners/baffles), a tail-gate to control the downstream level, a stilling tank and calibrated flow meter. Materials: concrete, marine plywood, acrylic (transparent, for flow visualisation), PVC, 3-D printed parts, cement mortar finishes (set to the roughness needed).
- **Step 6 – Instrumentation:** point and hook gauges (water level), Pitot tubes, ADV, propeller meters, pressure transducers/piezometers, load cells, PIV/LSPIV, dye tracers, high-speed video.
- **Step 7 – Verification (calibration):** reproduce a known prototype condition (rating curve, high-water marks, tide records) by adjusting roughness; then test the proposed design for different alternatives.
- **Step 8 – Testing and interpretation:** conduct runs, record data, convert to prototype units, apply scale-effect corrections, and provide recommendations.

10.2 Movable-bed models

The bed material must be scaled so the model simulates **incipient motion, bed forms and transport**. Conditions: equal Shields (mobility) parameter, equal Froude number, particle Reynolds number R_* above ~ 10–70. Because natural sand scaled by 1/100 would become cohesive silt, **lightweight materials** (crushed coal, bakelite, polystyrene, walnut shells, plastic pellets) are used, density $\rho_{s,m}$ ~ 1050–1500 kg/m^3. The time scale of bed changes is found by sediment-continuity (Exner) scaling and usually differs from the flow time scale (an extra **sediment time-scale factor**).

10.3 Example – design of a spillway model

Prototype: ogee spillway, crest length 60 m, design head 8 m, design flood 3000 m^3/s; laboratory flume width 1.2 m and maximum flow 0.25 m^3/s. Choose Froude scale.

- Discharge limit: for the full 60 m crest, $L_r = 1/40$ would need $Q_m = 3000/40^{2.5} = 0.296\ m^3/s >$ 0.25 and a 1.5 m wide model > 1.2 m flume. So build a **sectional model** of a 48 m strip of crest (1.2 m in the flume).
- Unit discharge: $q_p = 3000/60 = 50\ m^2/s$; $q_m = 50/40^{1.5} = 0.198\ m^2/s$; $Q_m = 0.198 \times 1.2 = 0.237\ m^3/s \le 0.25$ ✓.
- Model design head = 8/40 = 0.20 m (> 0.05 m ✓, so surface-tension effects are negligible).
- Time scale: $1/\sqrt{40} = 1/6.3$. Velocity scale 1/6.3.

10.5 Quick revision – formulae

Topic	Formula
Rectangular sharp-crested weir	$Q = (2/3)C_d\sqrt{(2g)}\ L\ H^{3/2}$
Francis	$Q = 1.84(L - 0.1nH)H^{3/2}$
V-notch	$Q = (8/15)C_d\sqrt{(2g)}\ \tan(\theta/2)\ H^{5/2}$
Broad-crested weir	$Q = 1.705\ C_d\ L\ H^{3/2}$
Critical depth on crest	$y_c = (2/3)H$
Parshall flume (ft, cfs)	$Q = 4\ W\ H_a^{1.522\ W^{0.026}}$
Froude model	$V_r = L_r^{1/2}$, $Q_r = L_r^{5/2}$, $T_r = L_r^{1/2}$, $F_r = L_r^3$
Reynolds model	$V_r = \nu_r/L_r$, $Q_r = \nu_r L_r$
Distorted model	$V_r = \sqrt{Y_r}$, $Q_r = X_r Y_r^{3/2}$, $T_r = X_r/\sqrt{Y_r}$

✎ **Exam Question:** Explain with suitable examples why distorted models are used in hydraulics. Derive the scale ratios for velocity, discharge and time for a distorted model. (10 marks)